

BFS Traversal

Step 1: Start at E

Legend

Complete

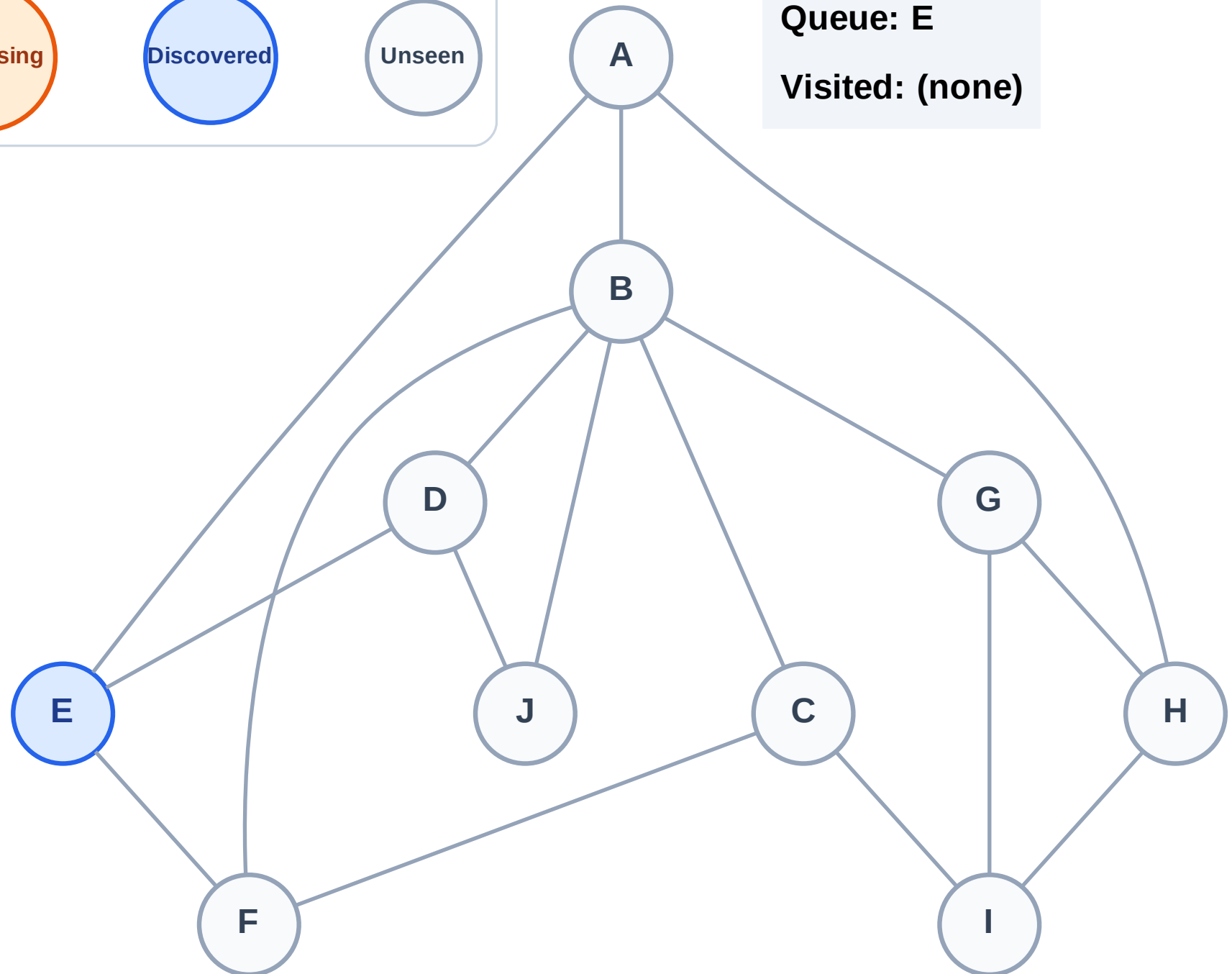
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

Complete

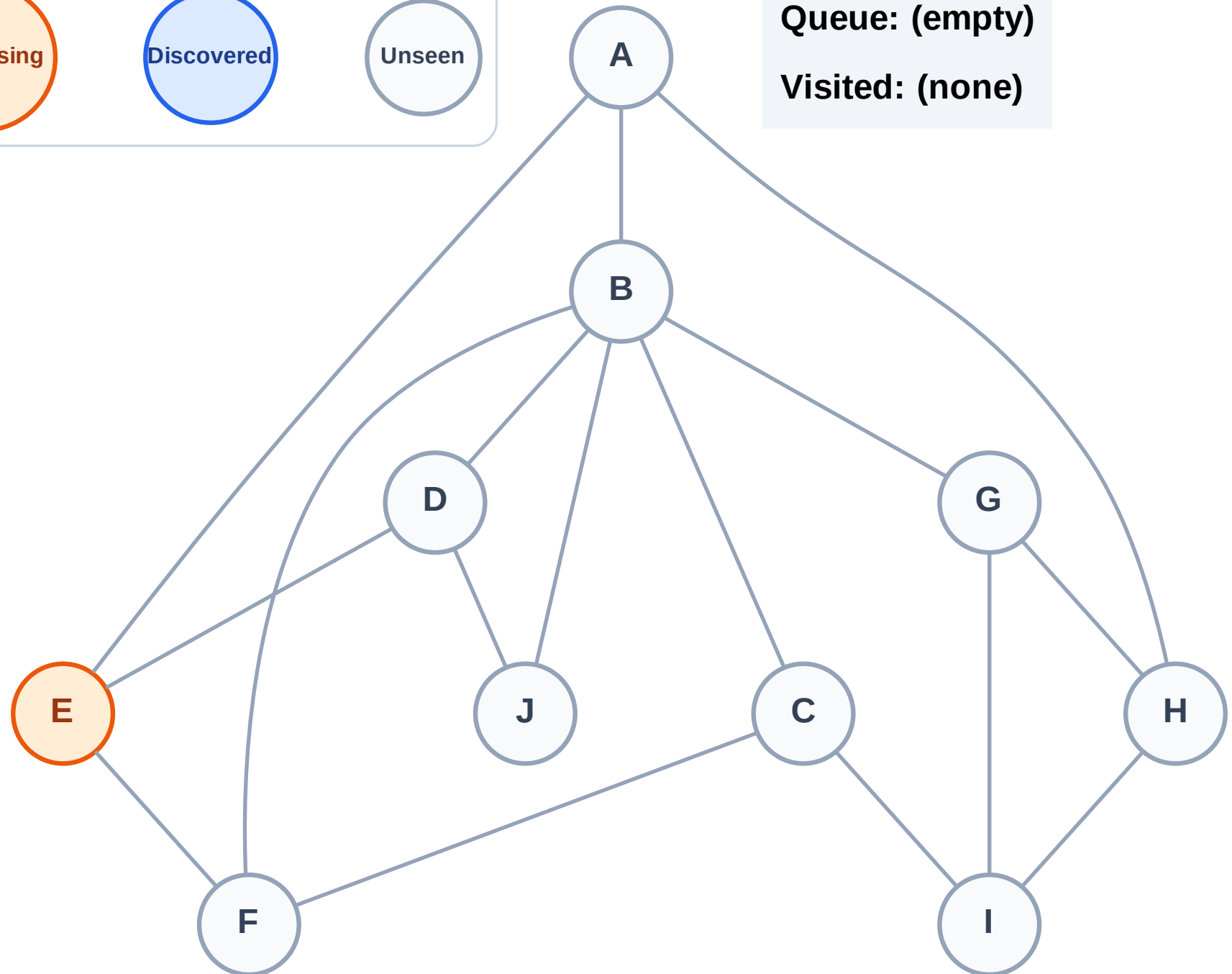
Processing

Discovered

Unseen

Queue: (empty)

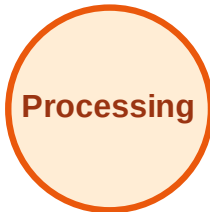
Visited: (none)



BFS Traversal

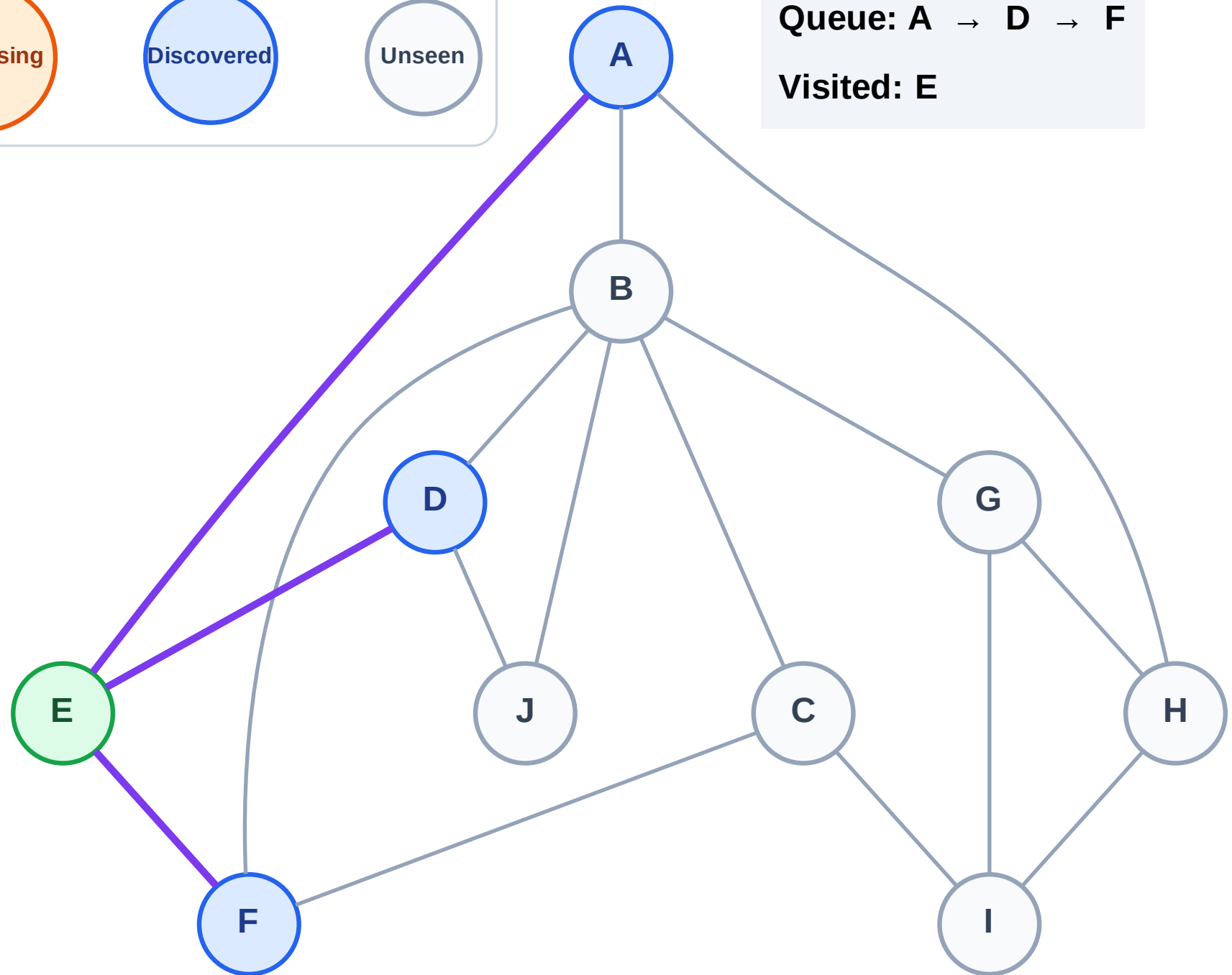
Step 3: Discover A, D, F from E

Legend



Queue: A → D → F

Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

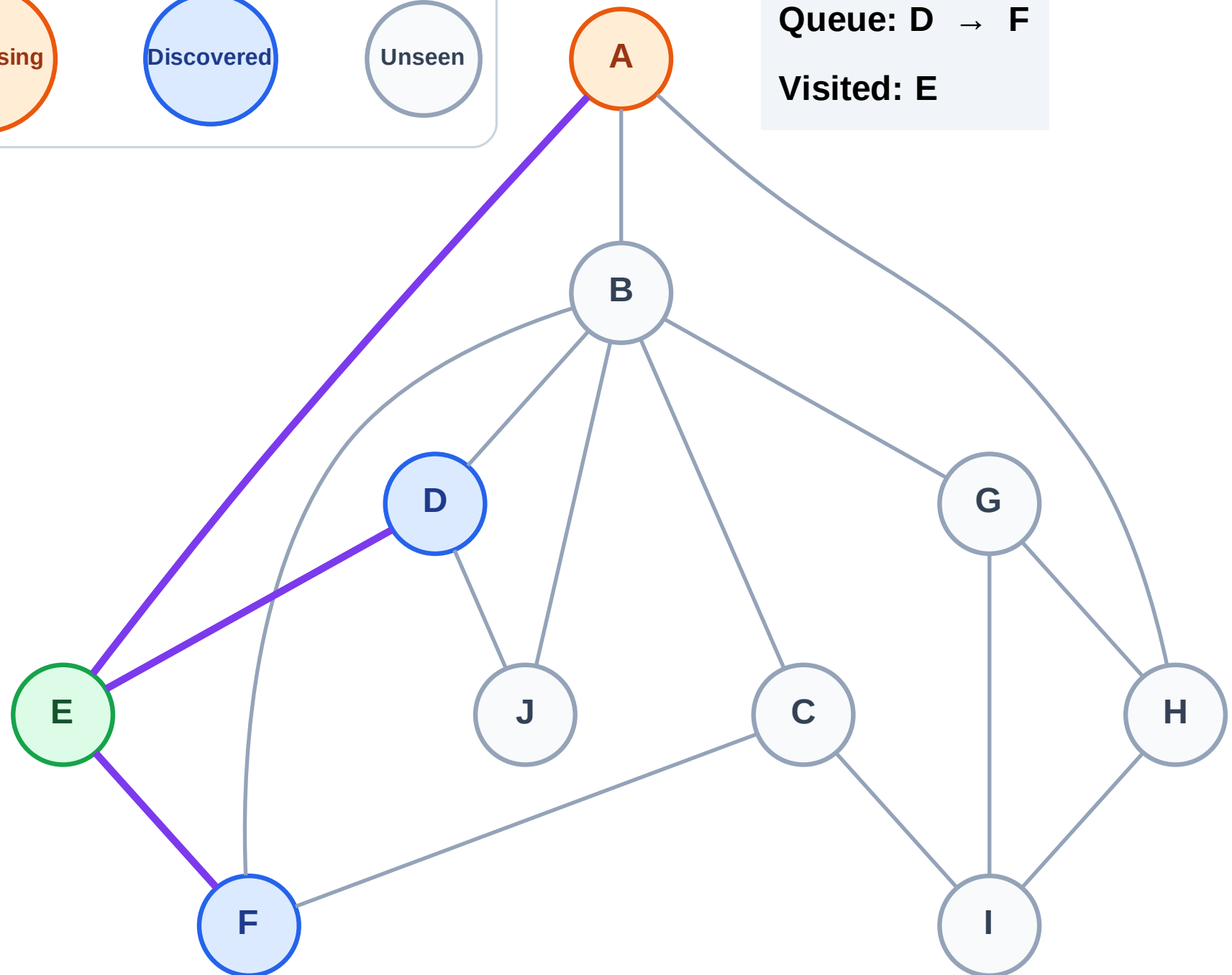
Processing

Discovered

Unseen

Queue: D → F

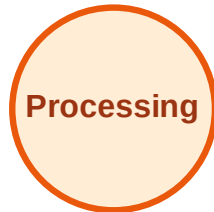
Visited: E



BFS Traversal

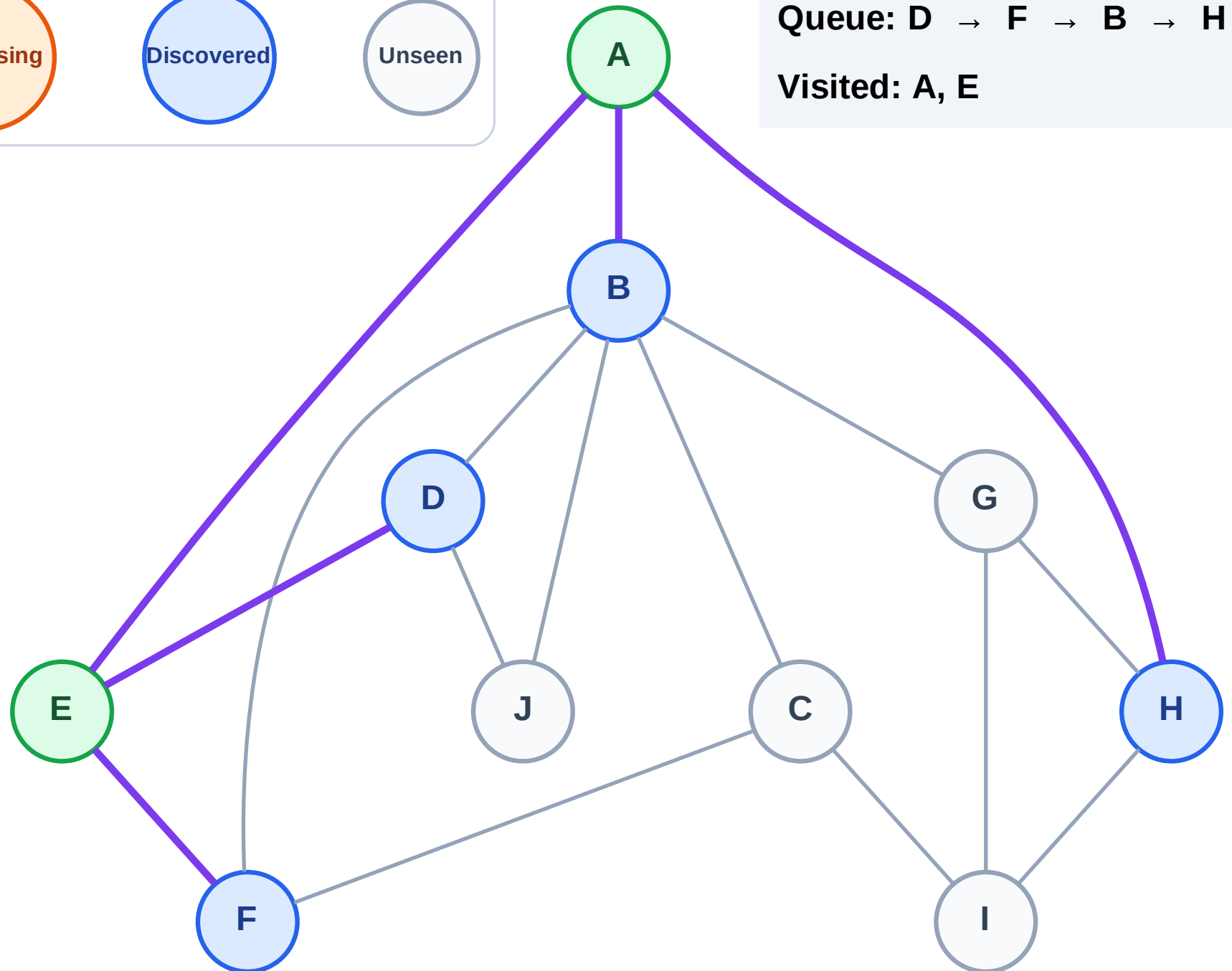
Step 5: Discover B, H from A

Legend



Queue: D → F → B → H

Visited: A, E



BFS Traversal

Step 6: Process node D

Legend

Complete

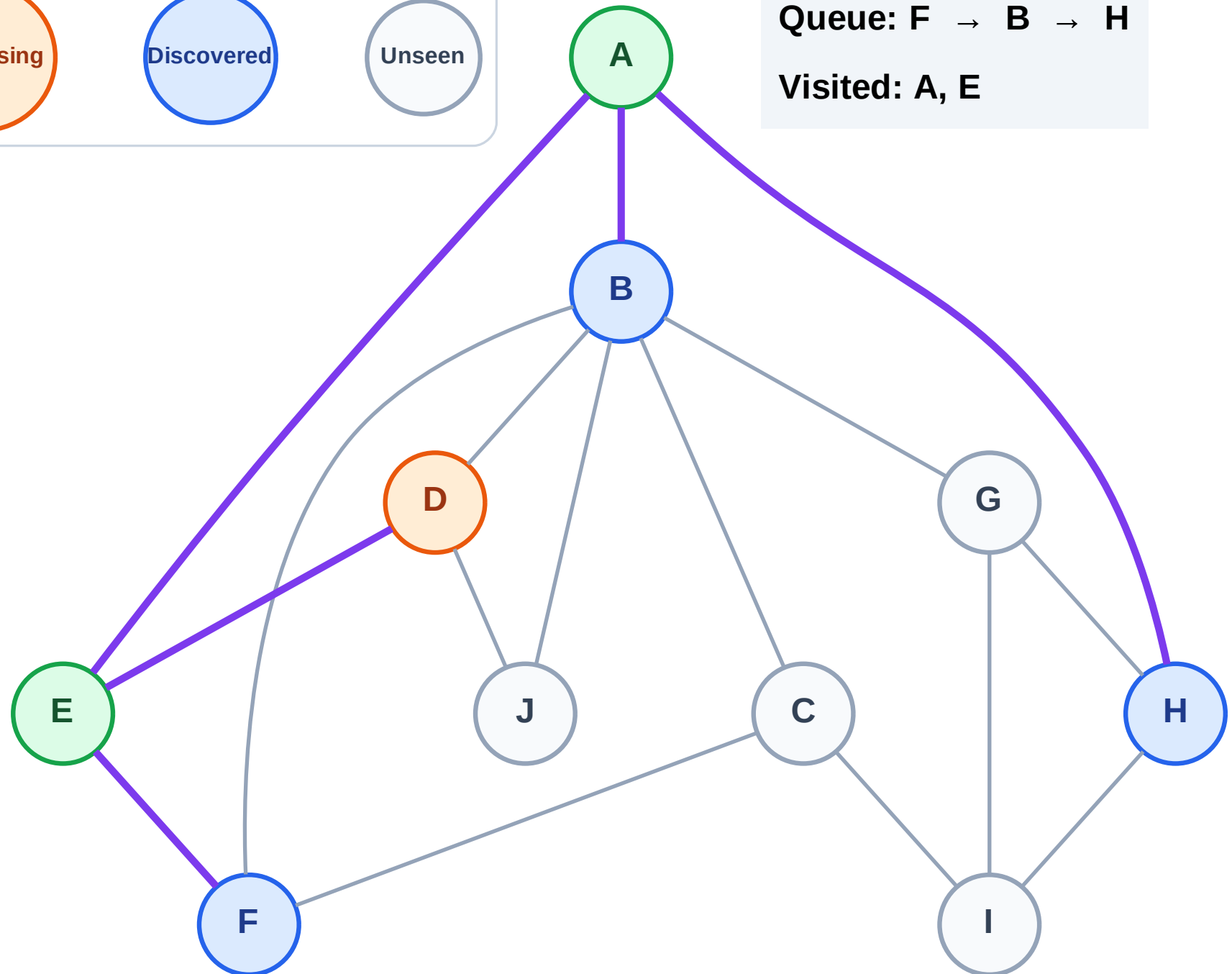
Processing

Discovered

Unseen

Queue: F → B → H

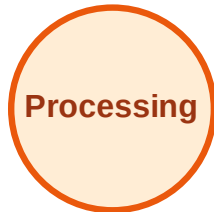
Visited: A, E



BFS Traversal

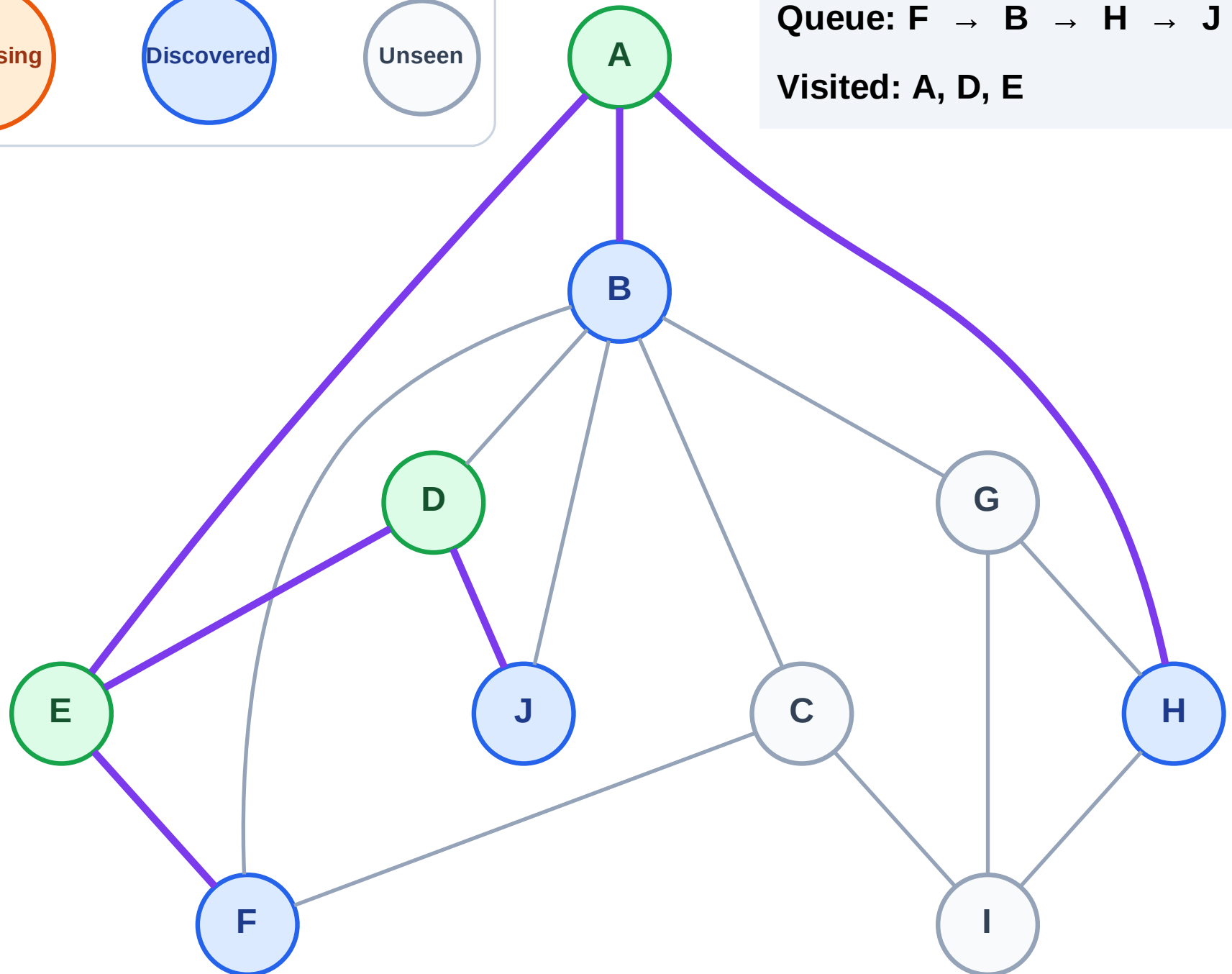
Step 7: Discover J from D

Legend



Queue: F → B → H → J

Visited: A, D, E



BFS Traversal

Step 8: Process node F

Legend

Complete

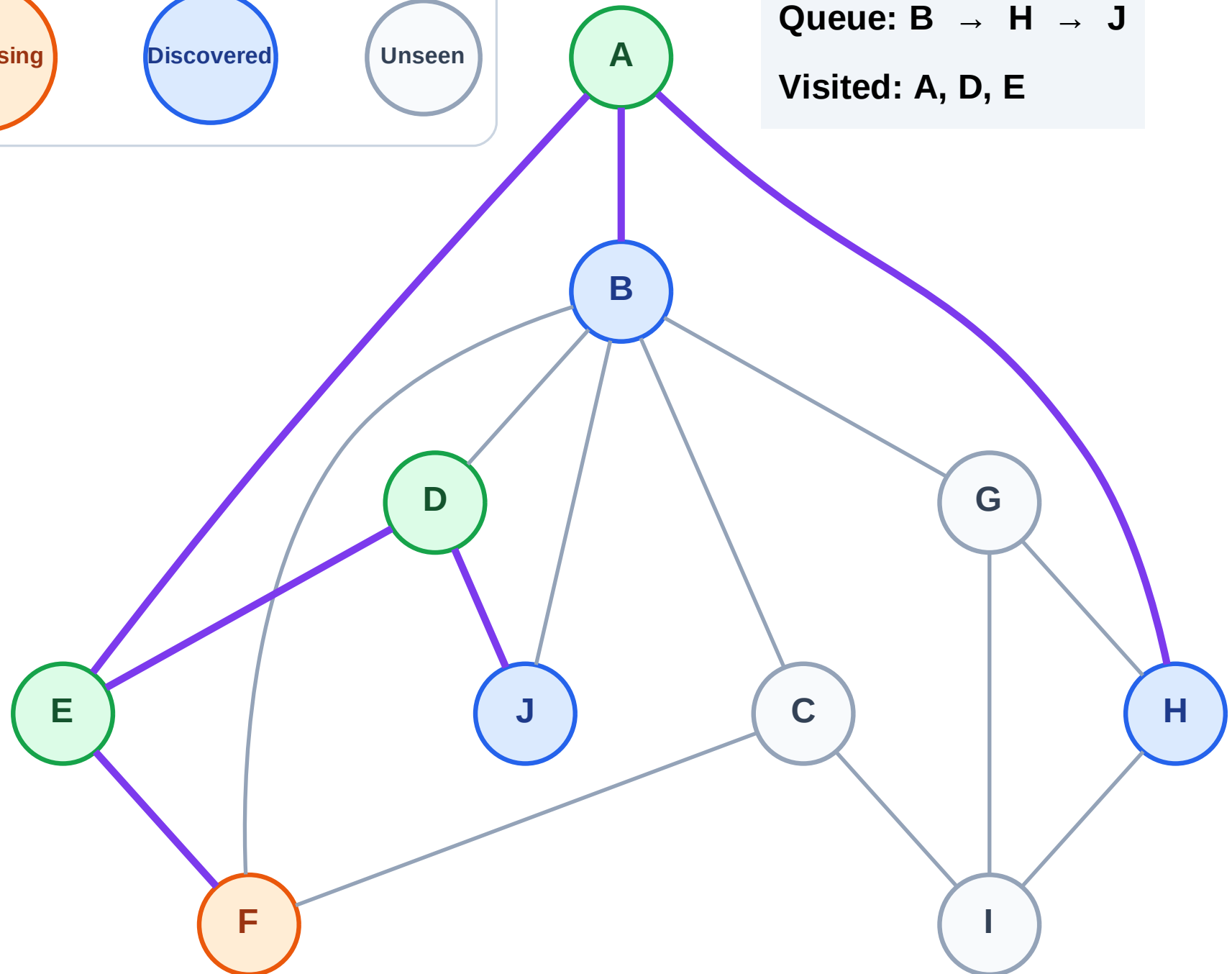
Processing

Discovered

Unseen

Queue: B → H → J

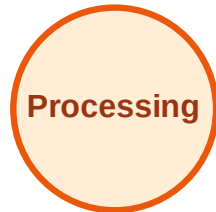
Visited: A, D, E



BFS Traversal

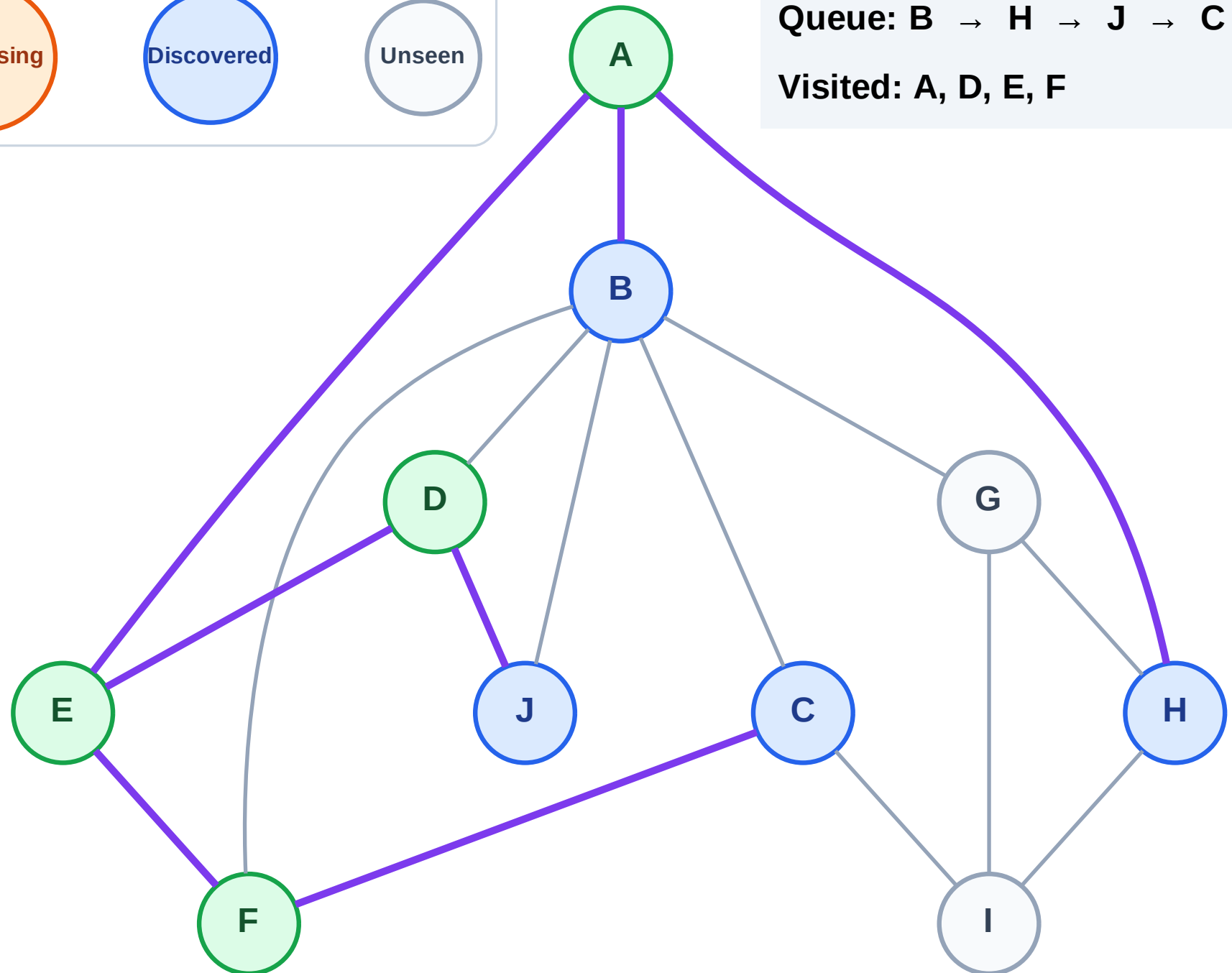
Step 9: Discover C from F

Legend



Queue: B → H → J → C

Visited: A, D, E, F



BFS Traversal

Step 10: Process node B

Legend

Complete

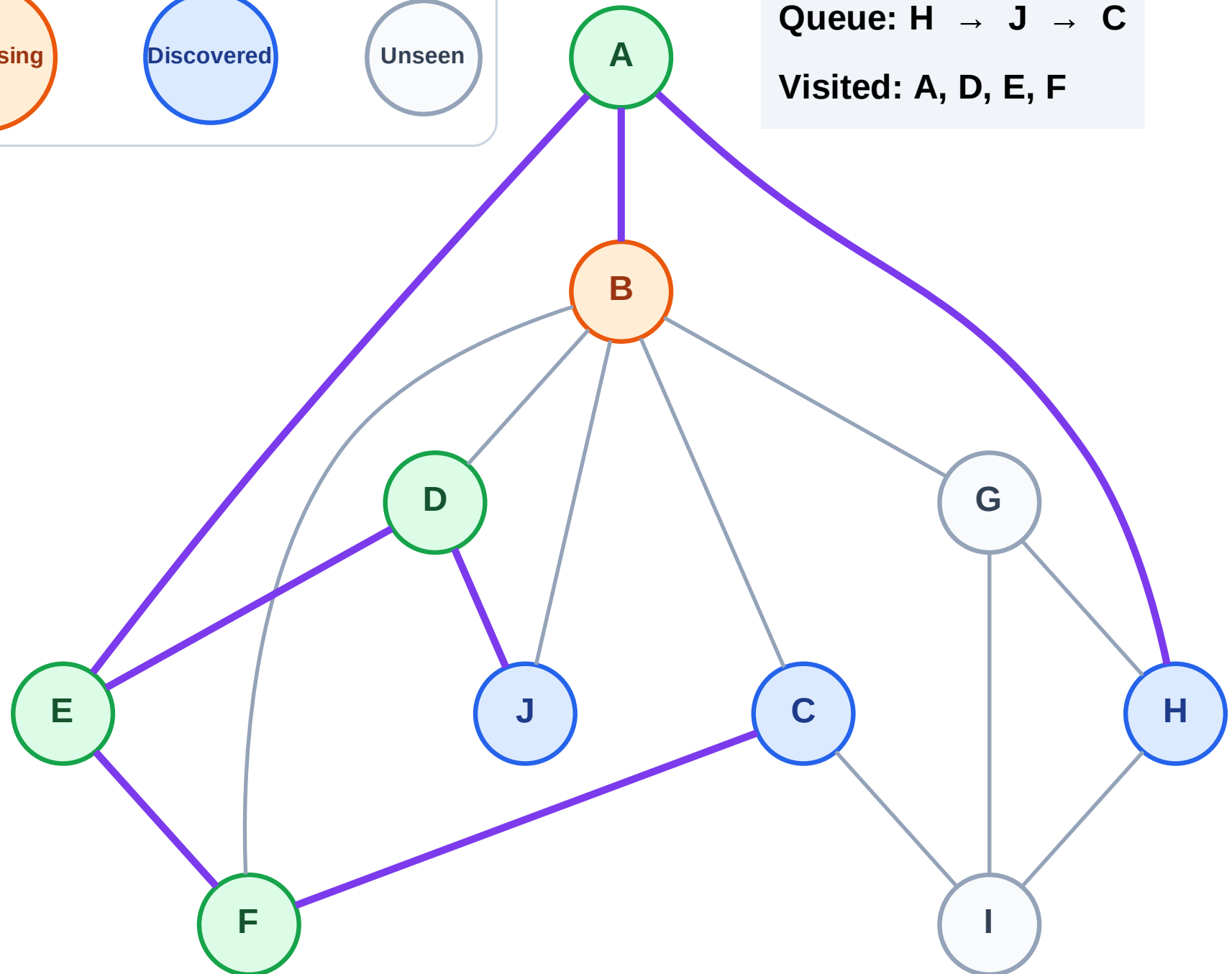
Processing

Discovered

Unseen

Queue: H → J → C

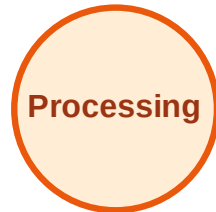
Visited: A, D, E, F



BFS Traversal

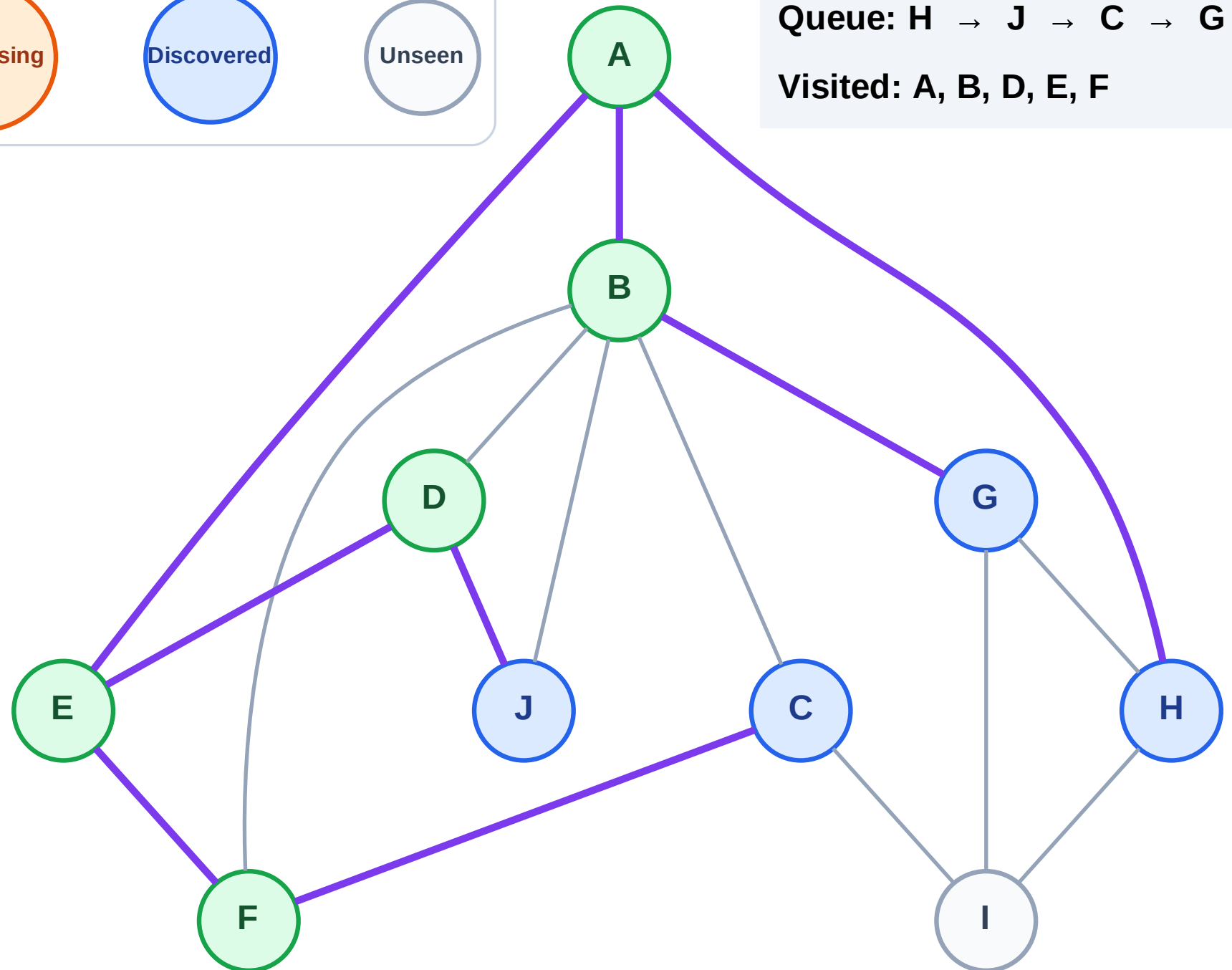
Step 11: Discover G from B

Legend



Queue: H → J → C → G

Visited: A, B, D, E, F



BFS Traversal

Step 12: Process node H

Legend

Complete

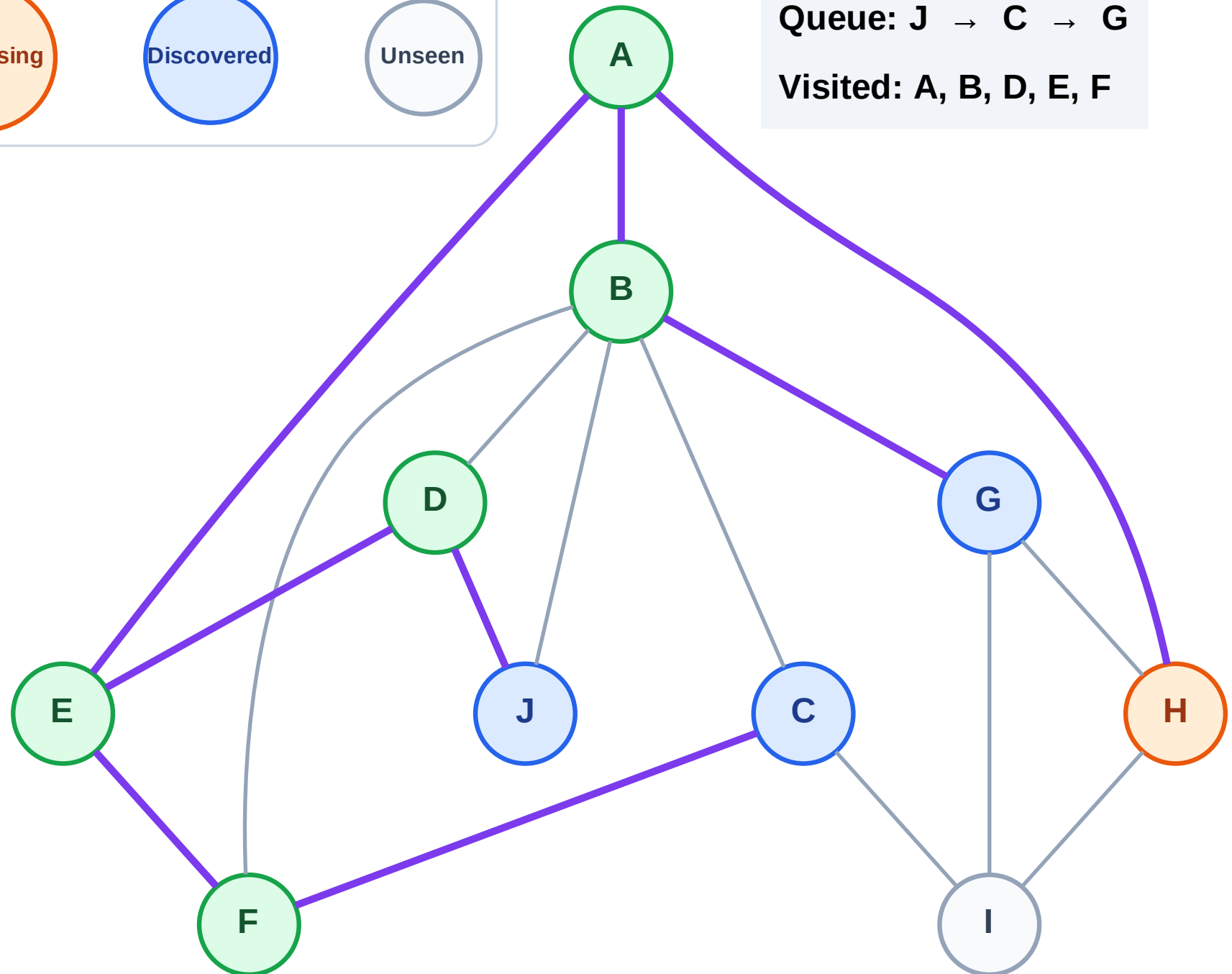
Processing

Discovered

Unseen

Queue: J → C → G

Visited: A, B, D, E, F



BFS Traversal

Step 13: Discover I from H

Legend

Complete

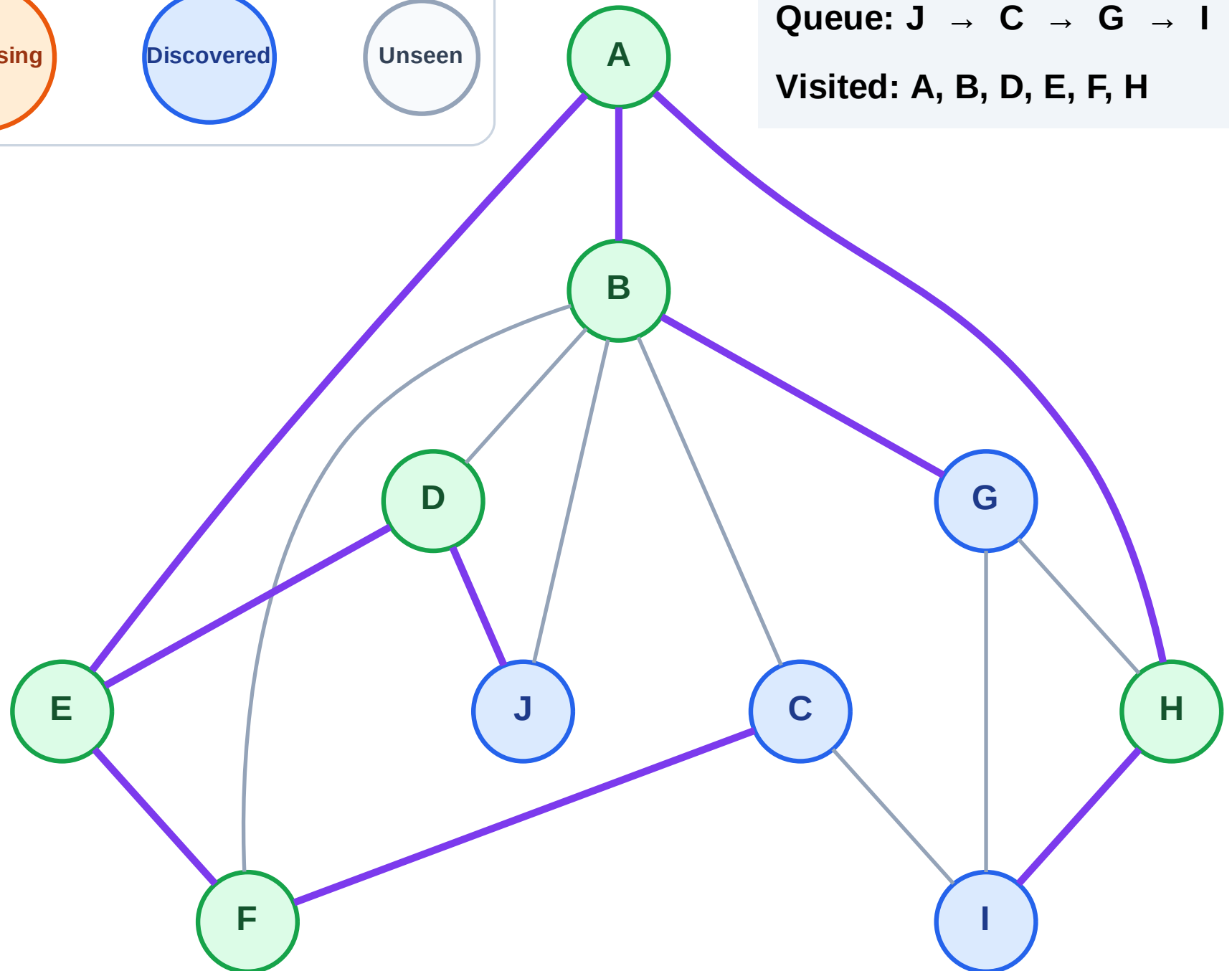
Processing

Discovered

Unseen

Queue: J → C → G → I

Visited: A, B, D, E, F, H



BFS Traversal

Step 14: Process node J

Legend

Complete

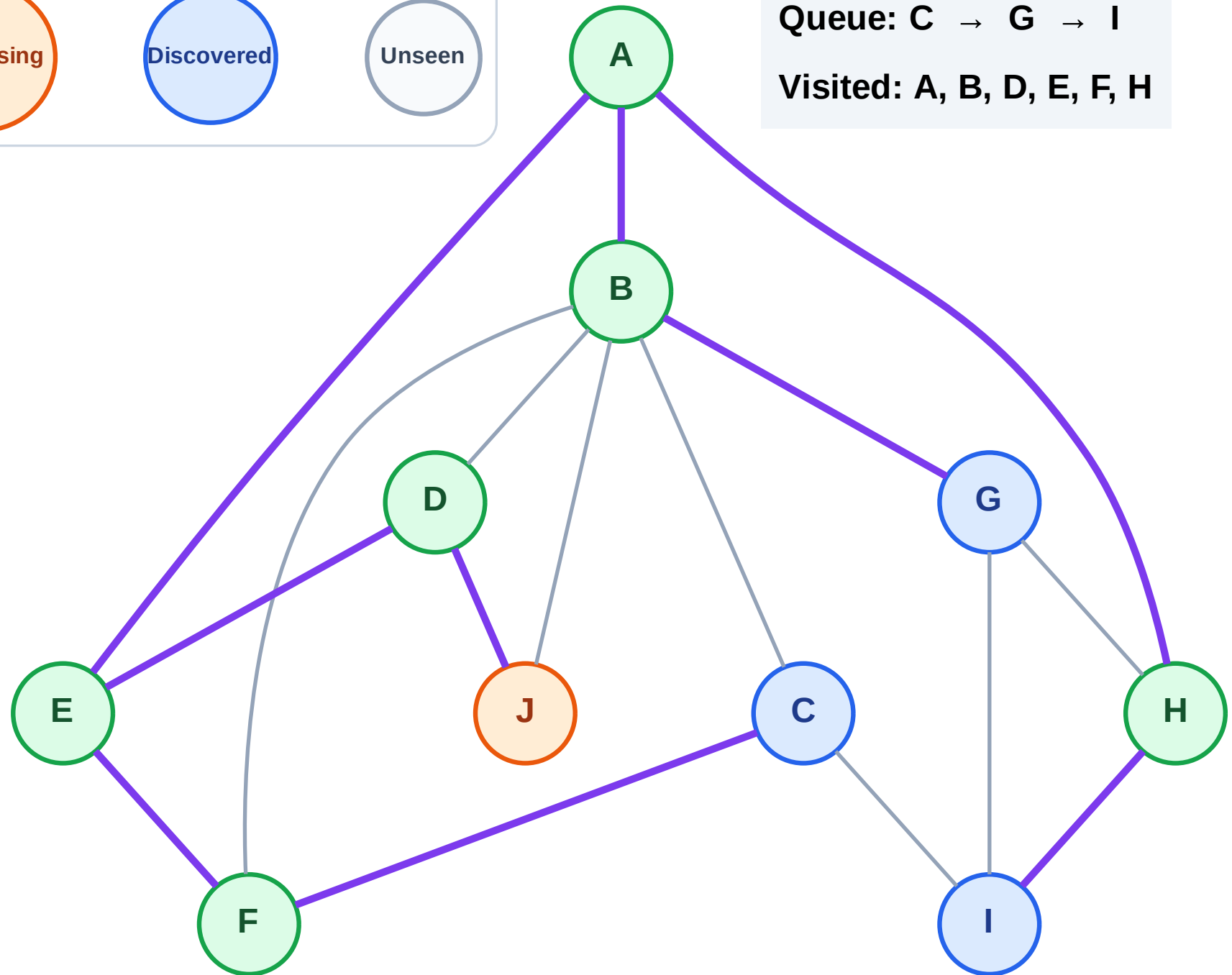
Processing

Discovered

Unseen

Queue: C → G → I

Visited: A, B, D, E, F, H



BFS Traversal

Step 15: No new neighbors from J

Legend

Complete

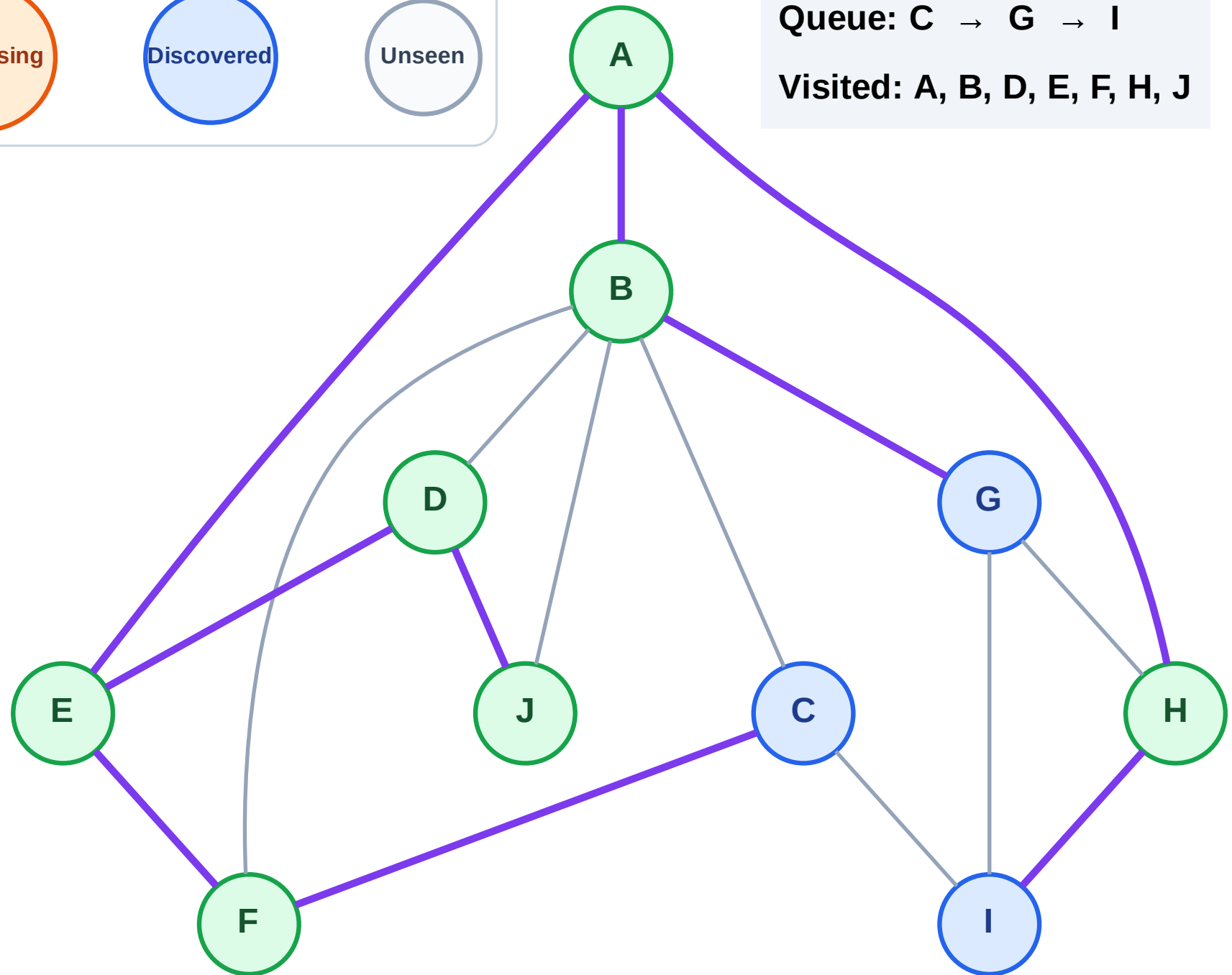
Processing

Discovered

Unseen

Queue: C → G → I

Visited: A, B, D, E, F, H, J



BFS Traversal

Step 16: Process node C

Legend

Complete

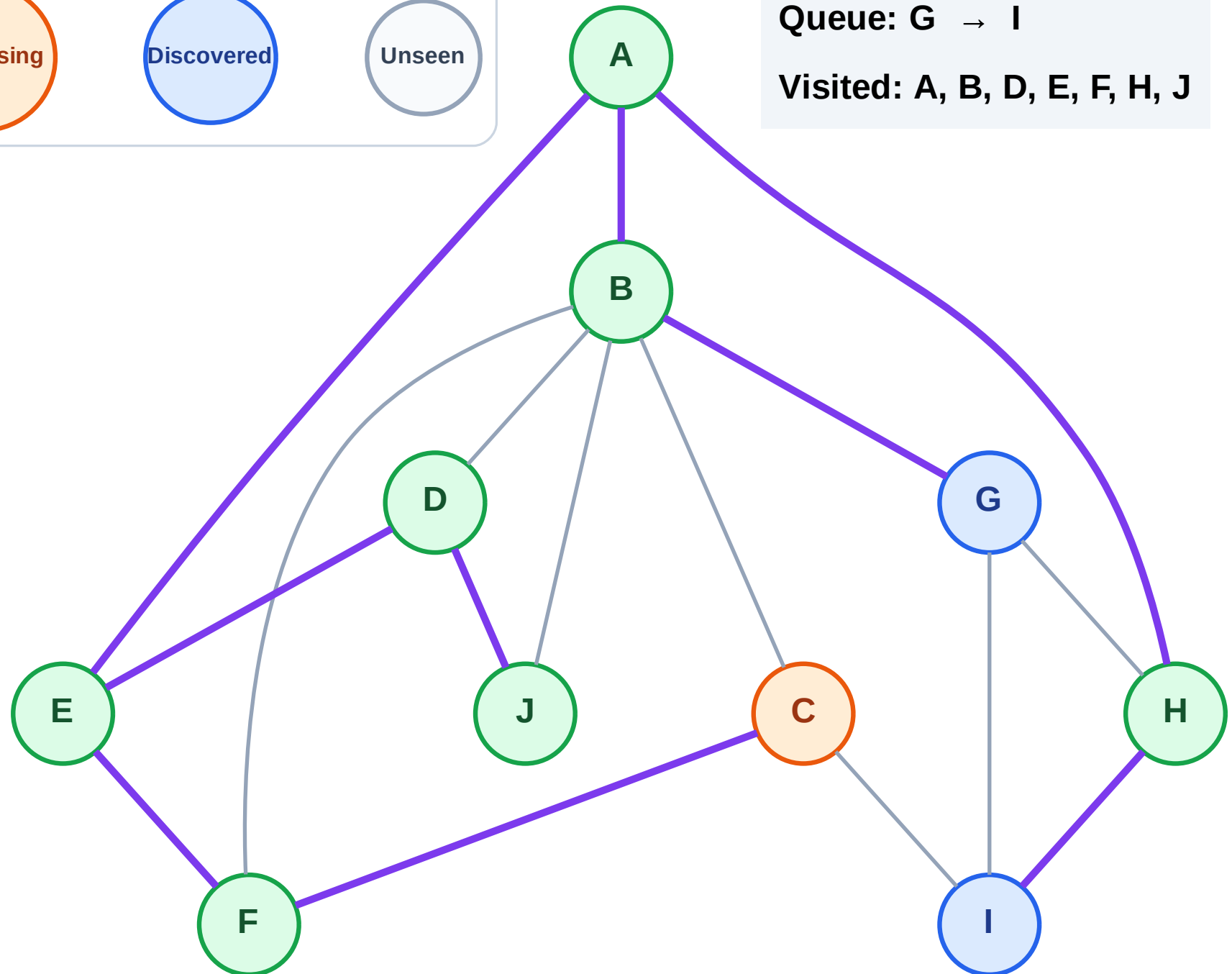
Processing

Discovered

Unseen

Queue: G → I

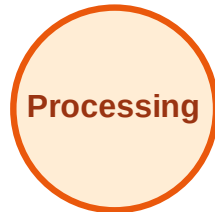
Visited: A, B, D, E, F, H, J



BFS Traversal

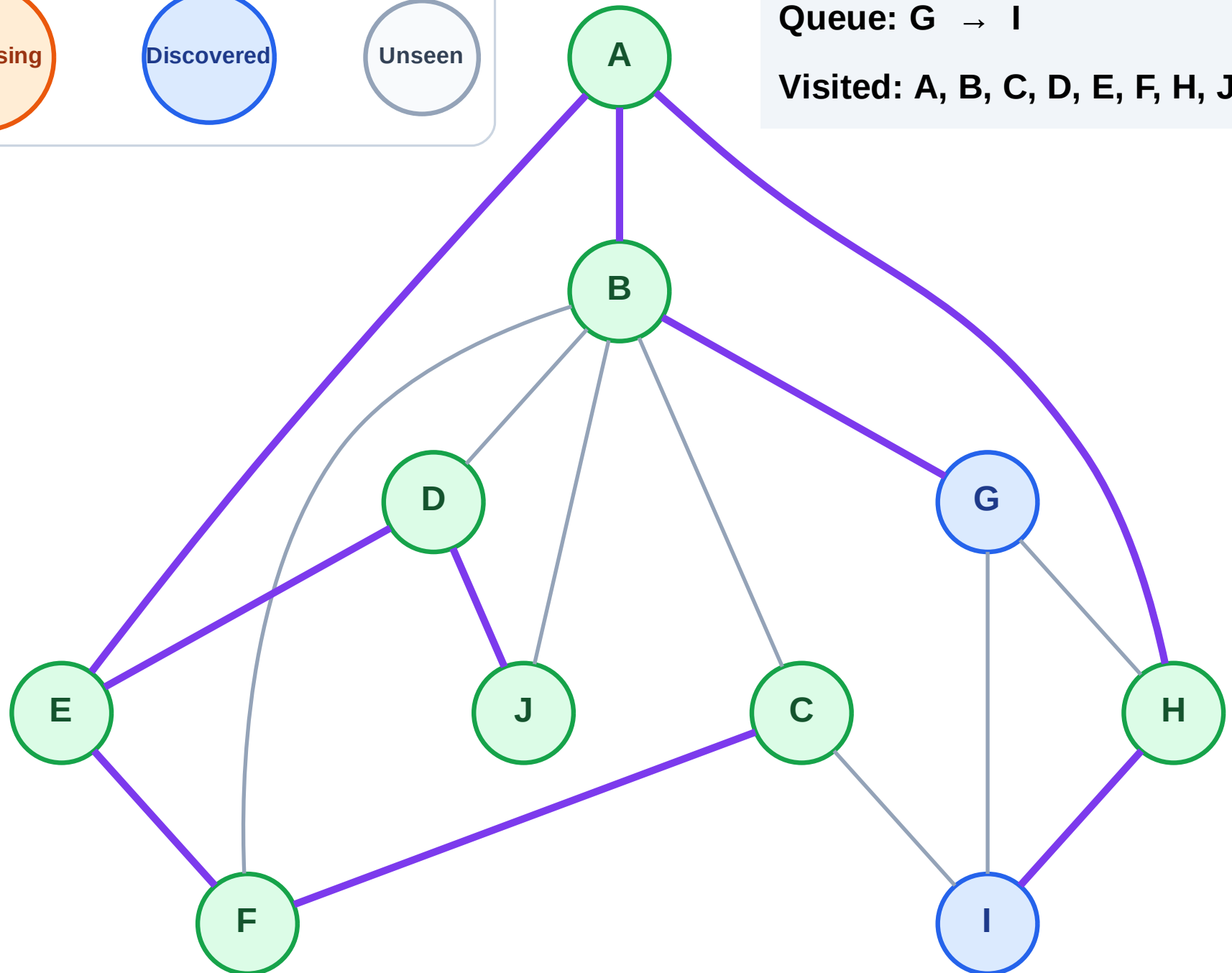
Step 17: No new neighbors from C

Legend



Queue: G → I

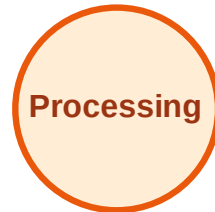
Visited: A, B, C, D, E, F, H, J



BFS Traversal

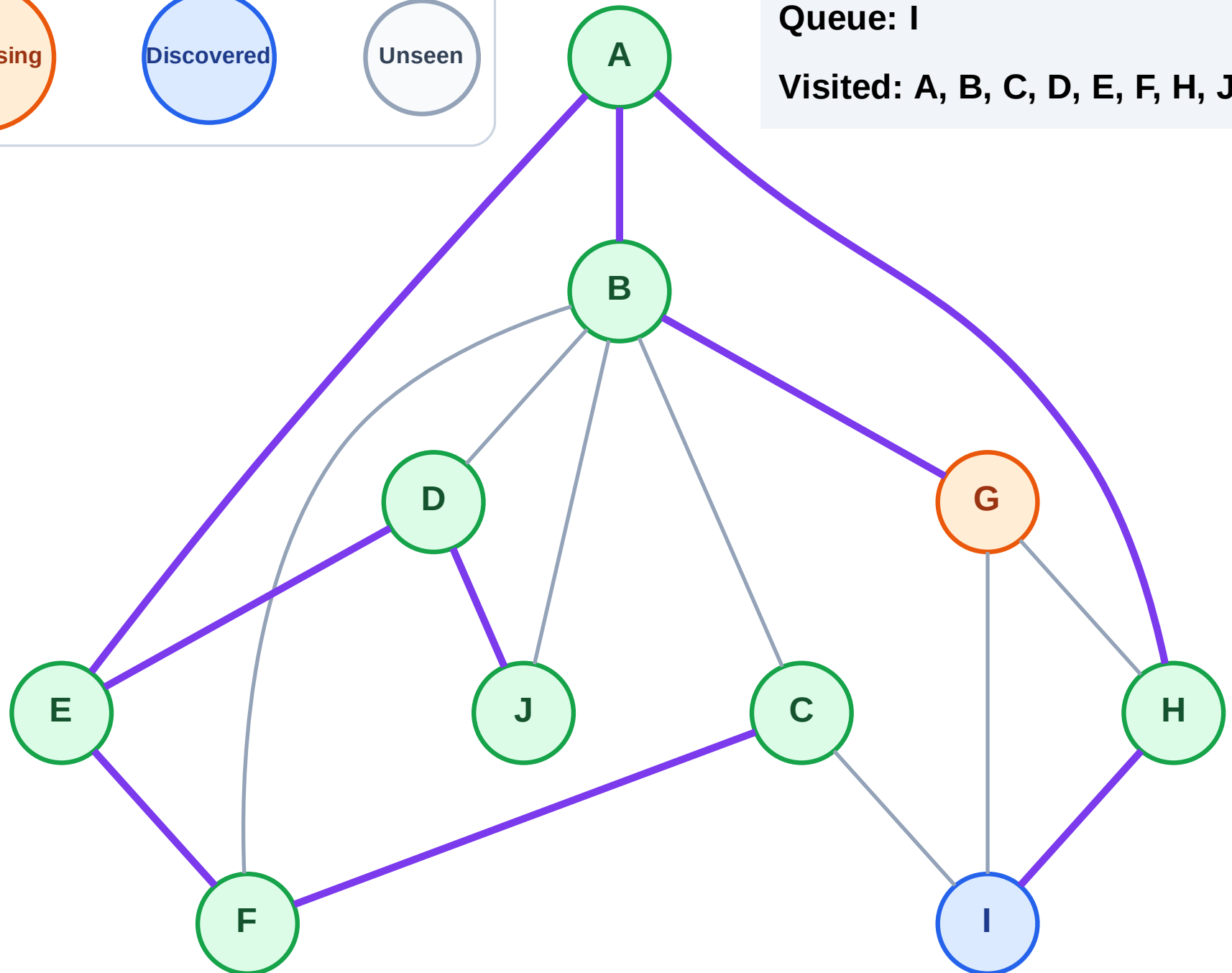
Step 18: Process node G

Legend



Queue: I

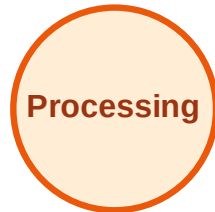
Visited: A, B, C, D, E, F, H, J



BFS Traversal

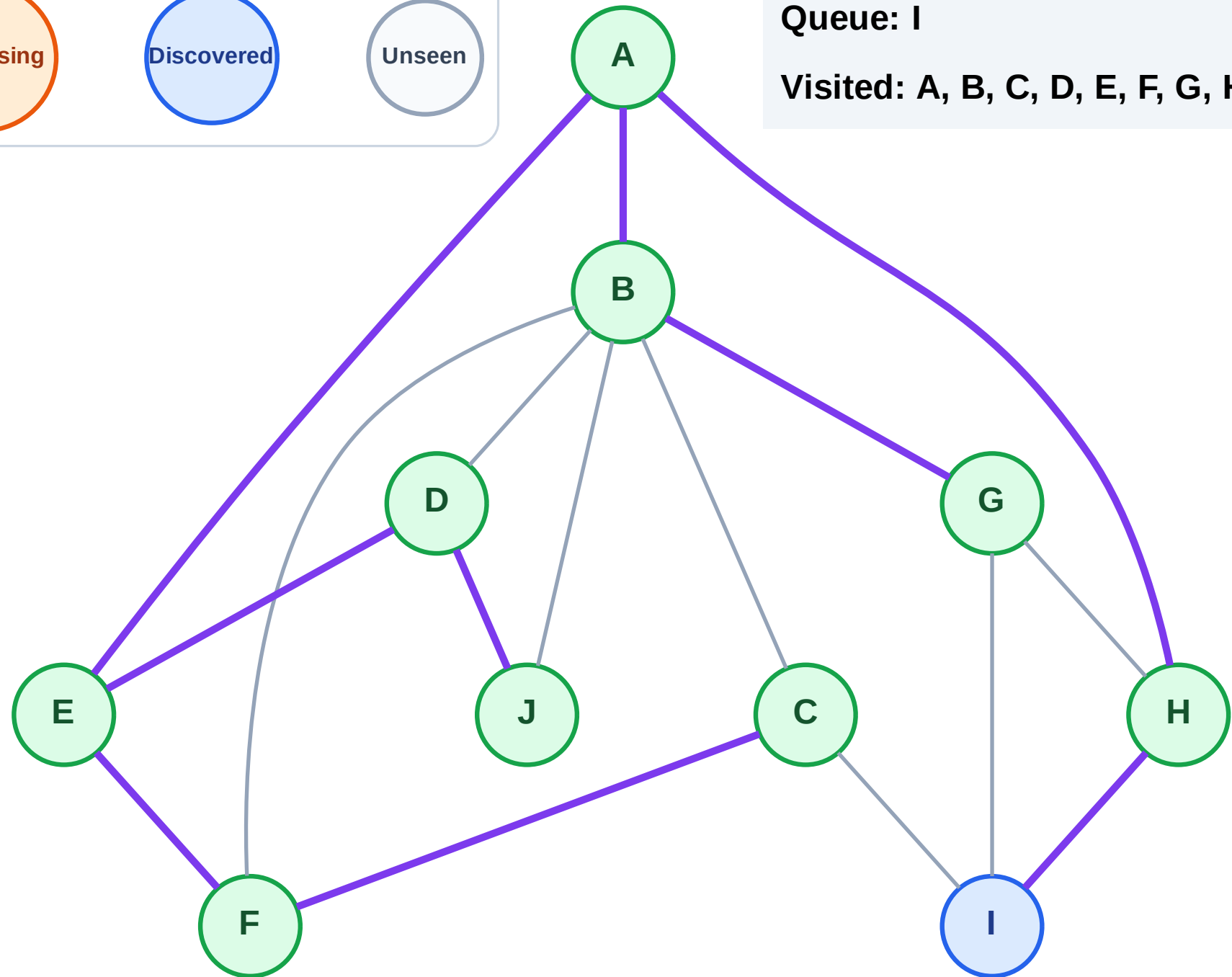
Step 19: No new neighbors from G

Legend



Queue: I

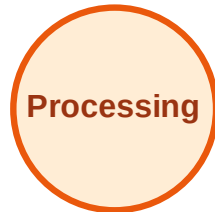
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

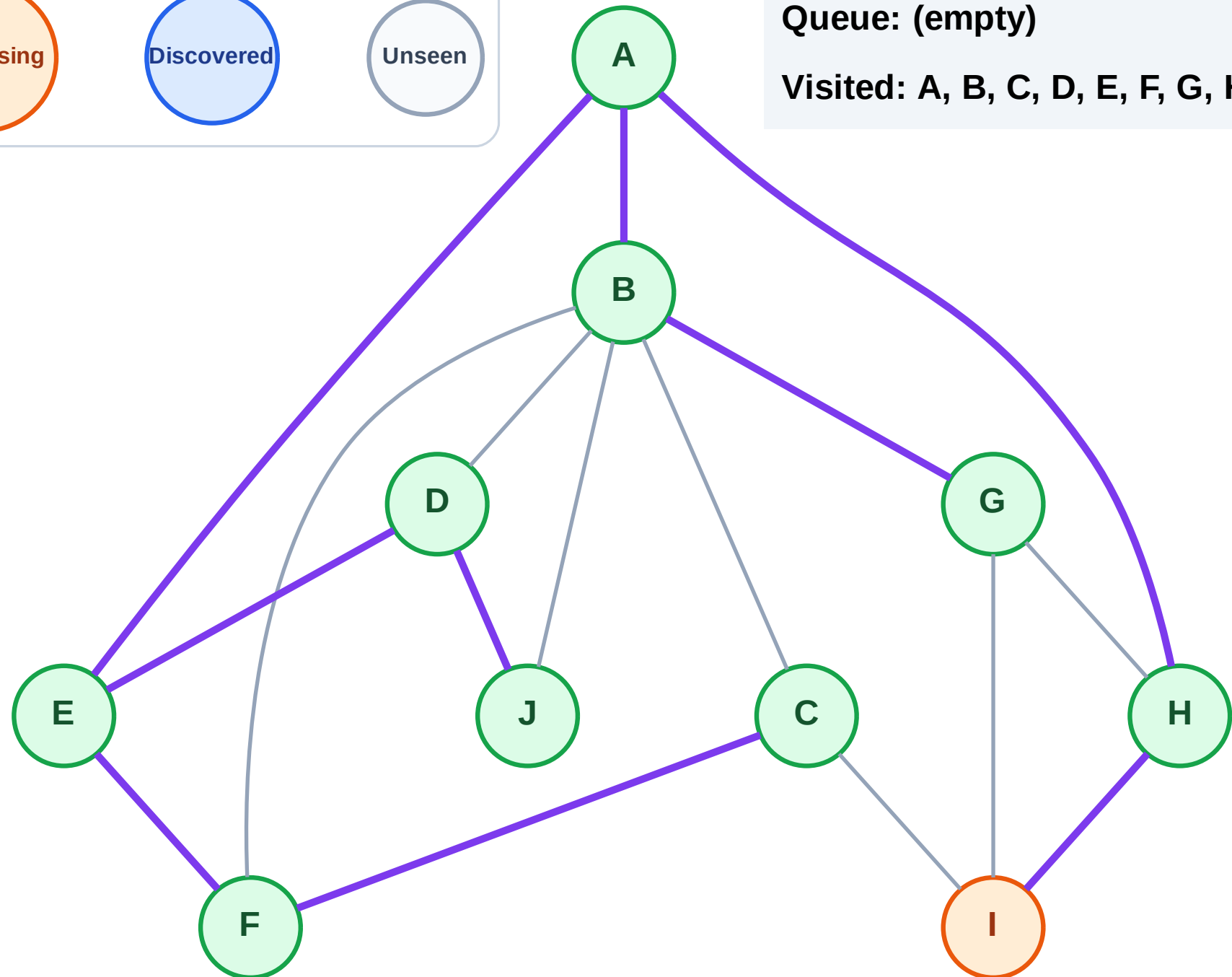
Step 20: Process node I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J

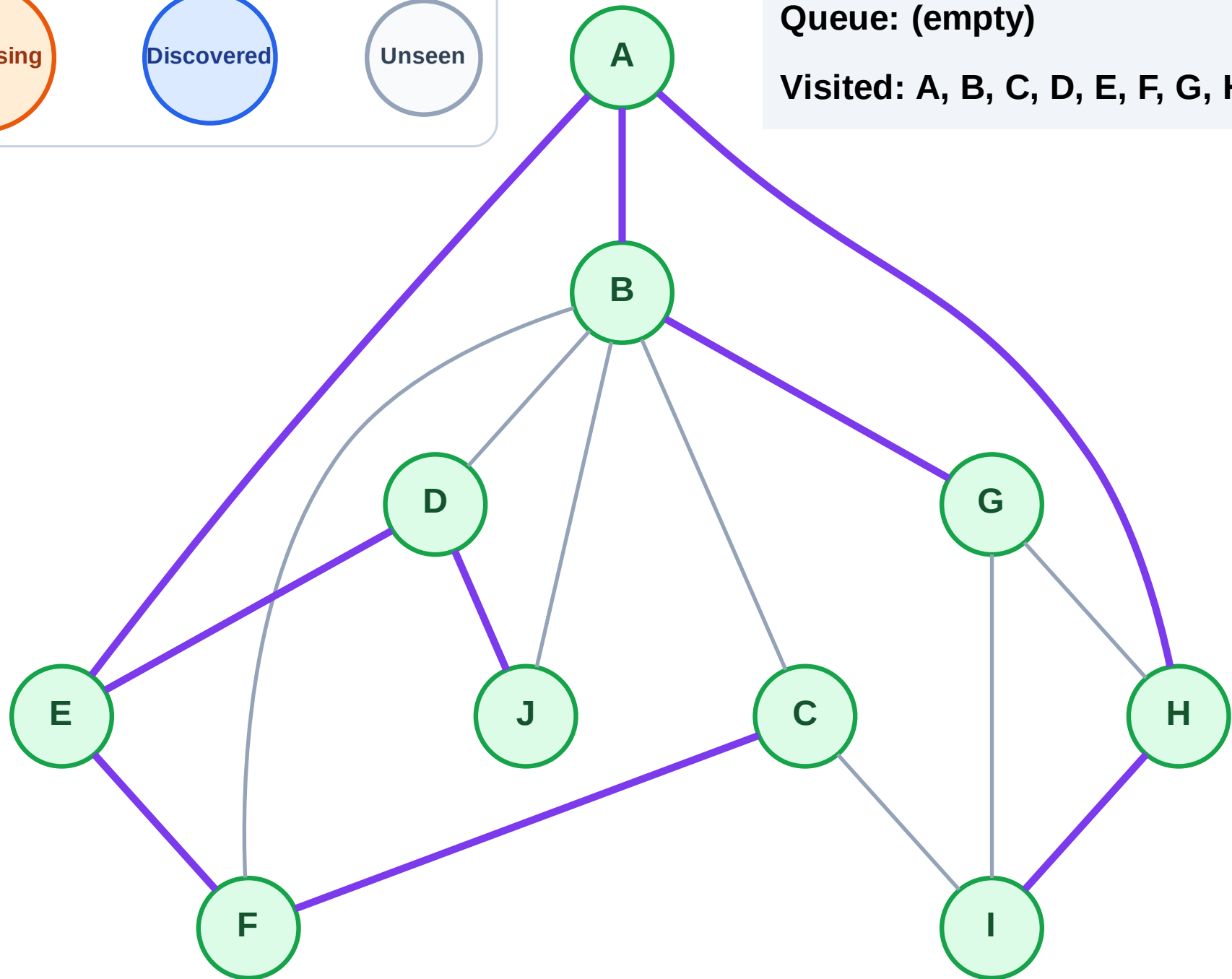


Step 21: No new neighbors from I

Legend



Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

Complete

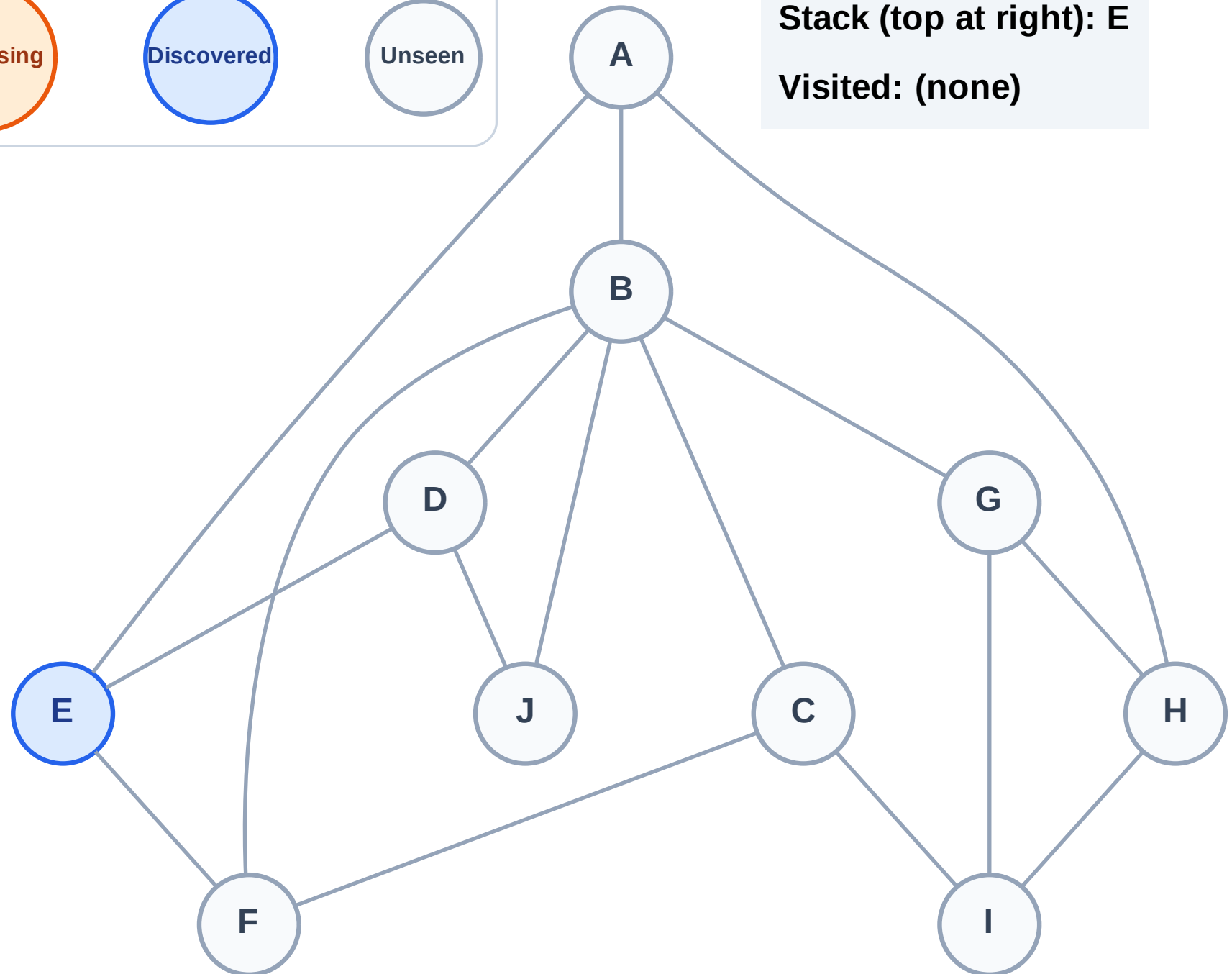
Processing

Discovered

Unseen

Stack (top at right): E

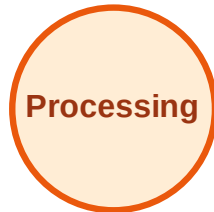
Visited: (none)



DFS Traversal

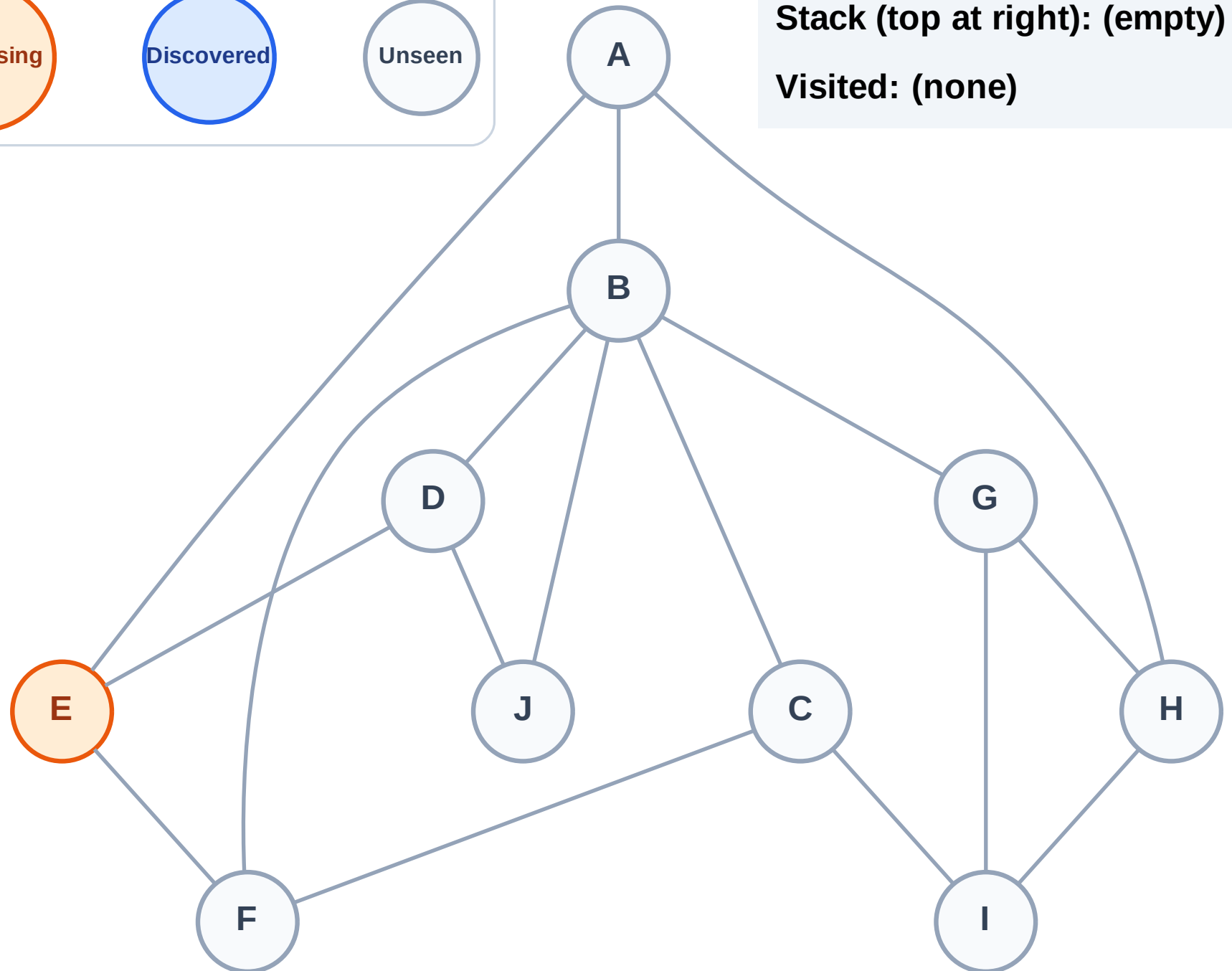
Step 2: Process node E

Legend



Stack (top at right): (empty)

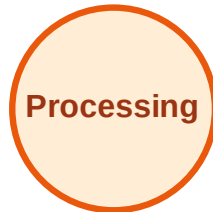
Visited: (none)



DFS Traversal

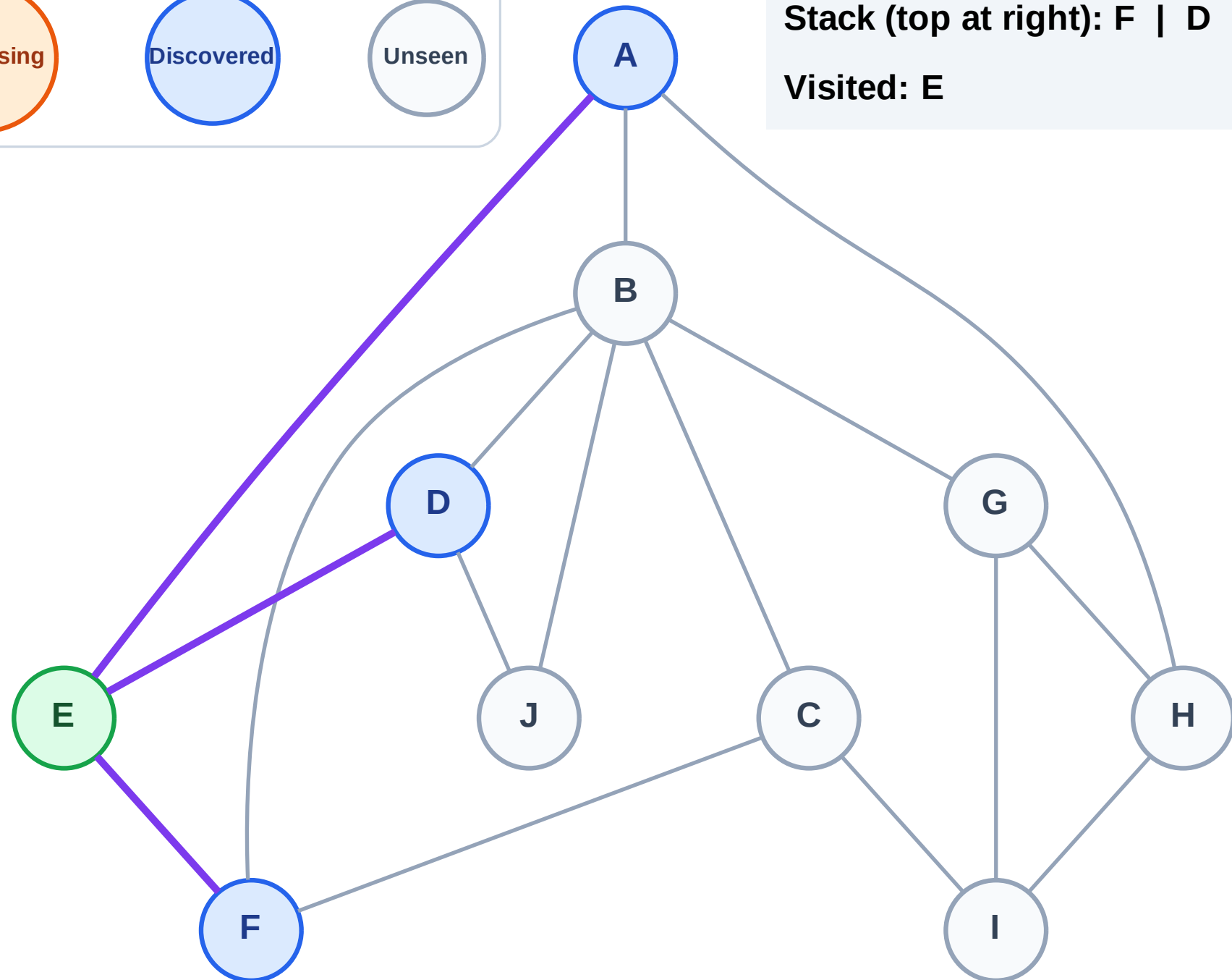
Step 3: Discover F, D, A from E

Legend



Stack (top at right): F | D | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

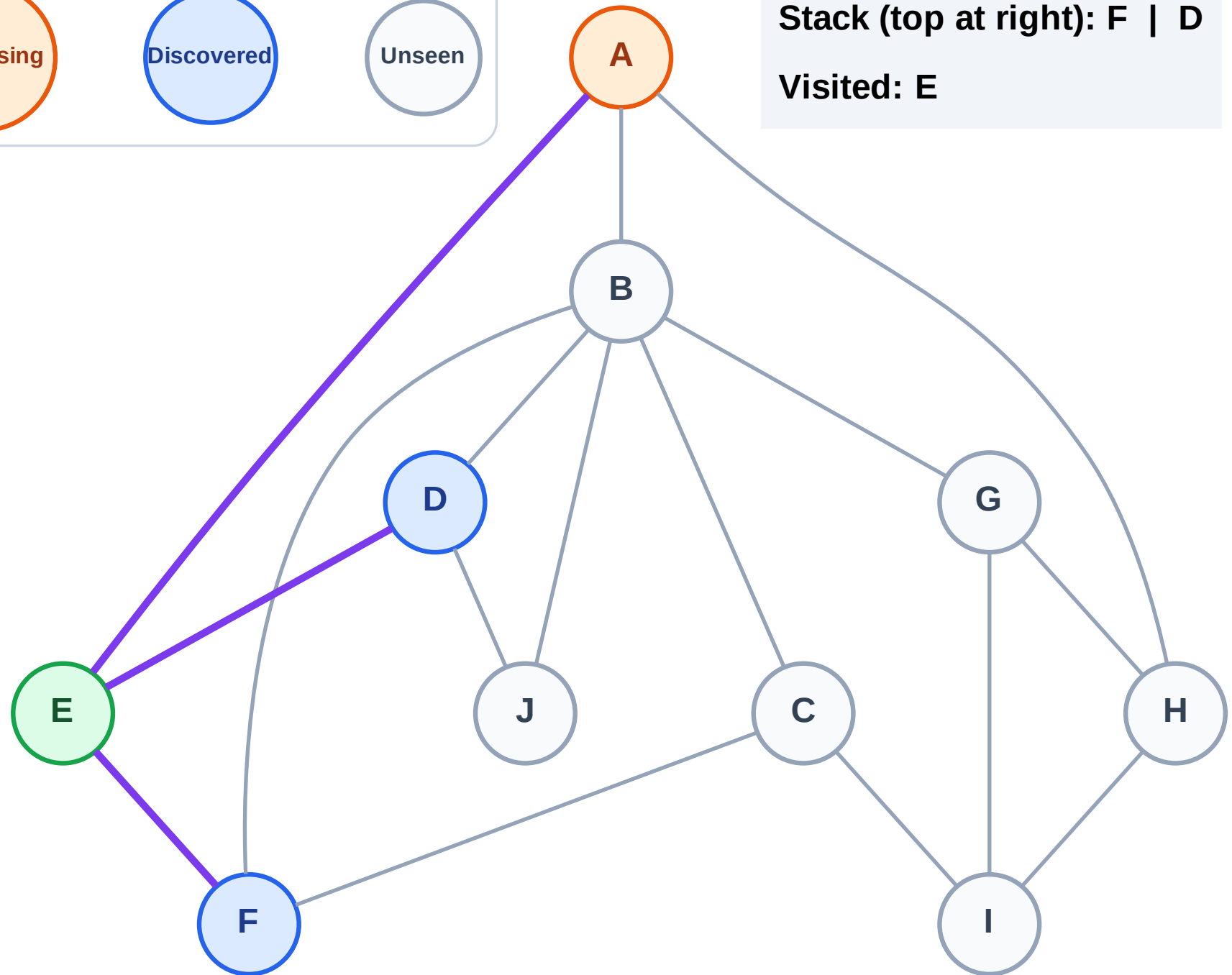
Processing

Discovered

Unseen

Stack (top at right): F | D

Visited: E



DFS Traversal

Step 5: Discover H, B from A

Legend

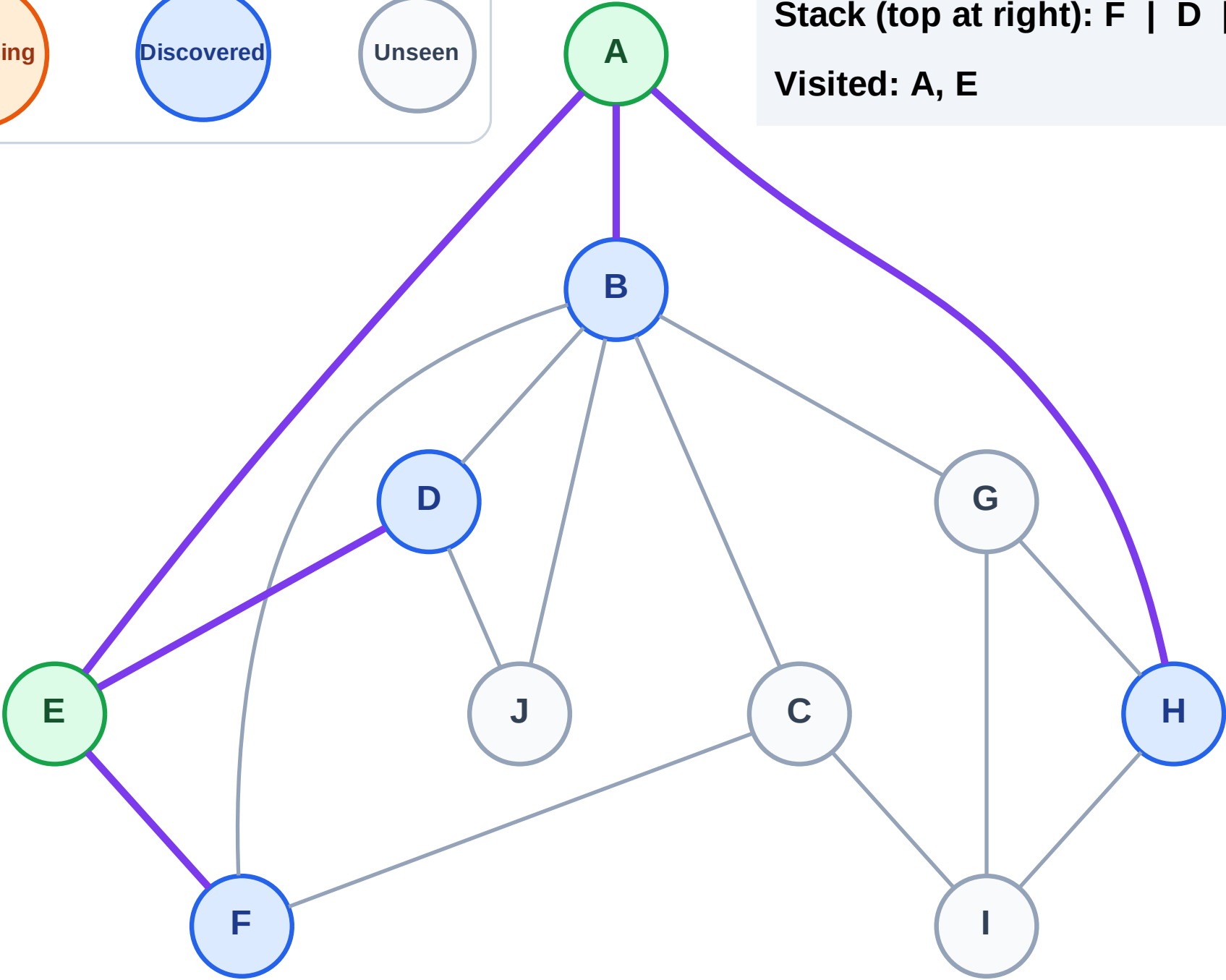
Complete

Processing

Discovered

Unseen

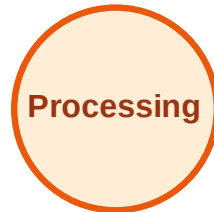
Stack (top at right): F | D | H | B
Visited: A, E



DFS Traversal

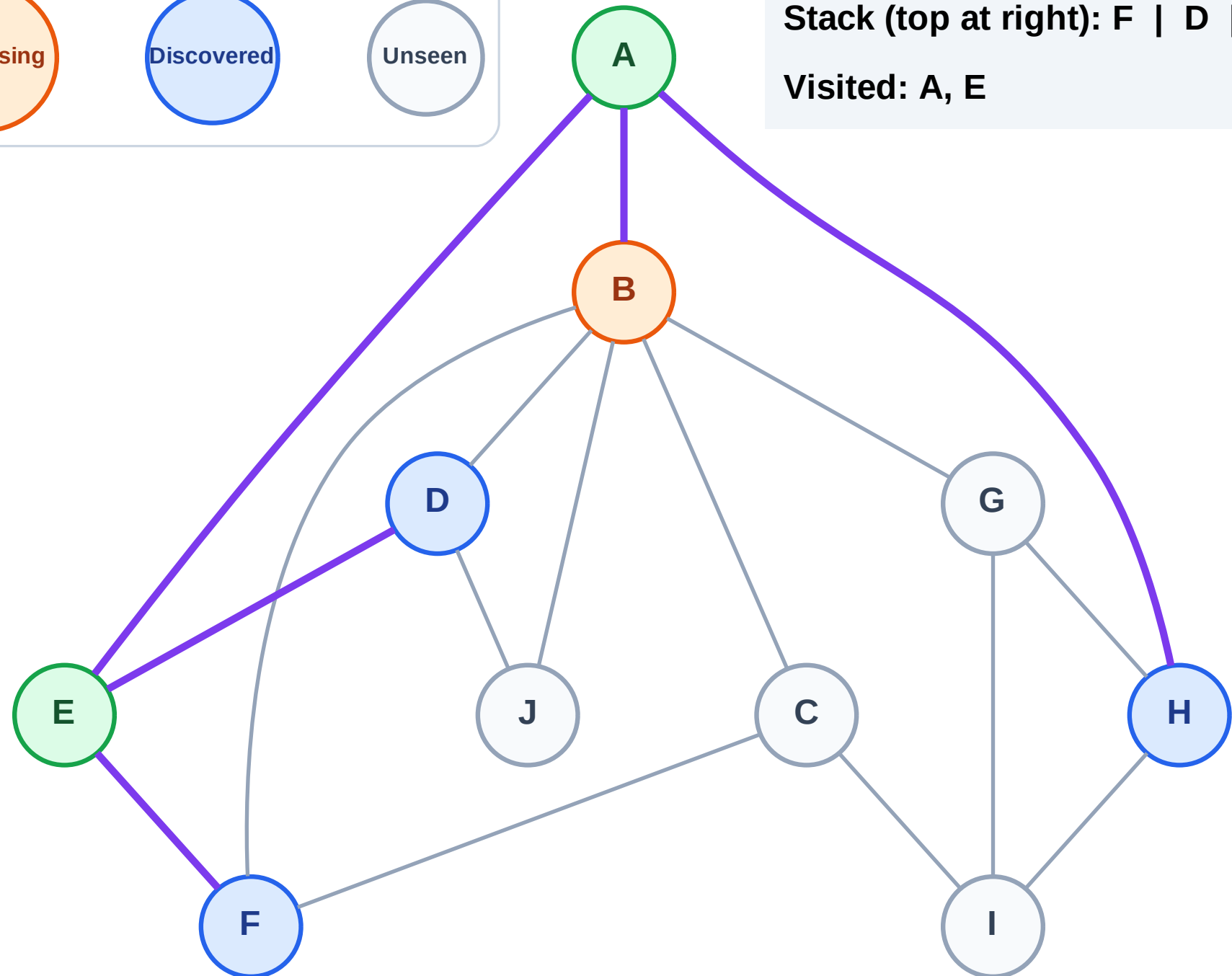
Step 6: Process node B

Legend



Stack (top at right): F | D | H

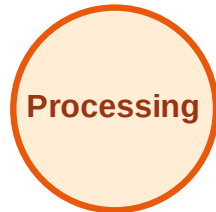
Visited: A, E



DFS Traversal

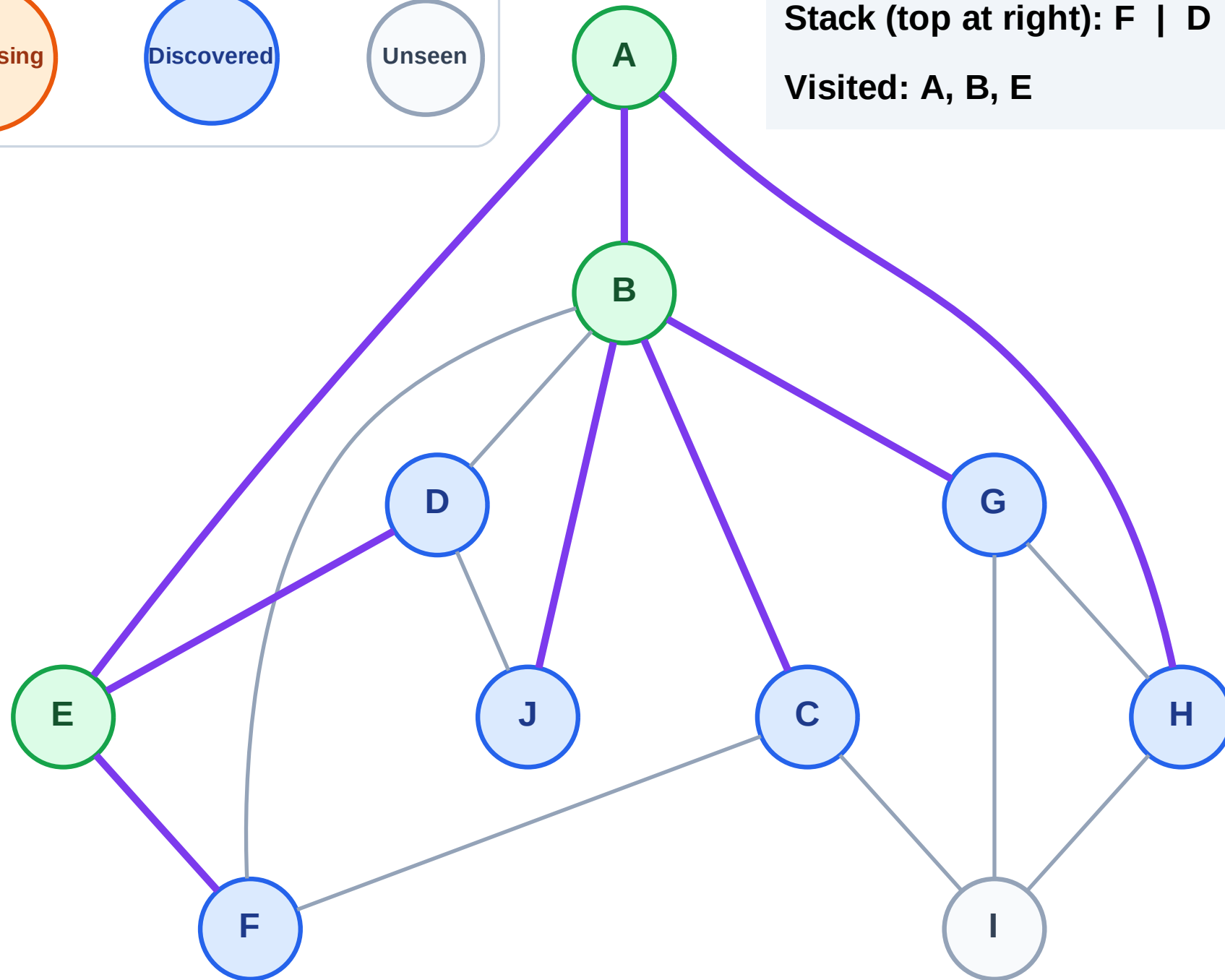
Step 7: Discover J, G, C from B

Legend



Stack (top at right): F | D | H | J | G | C

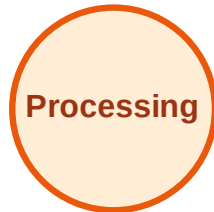
Visited: A, B, E



DFS Traversal

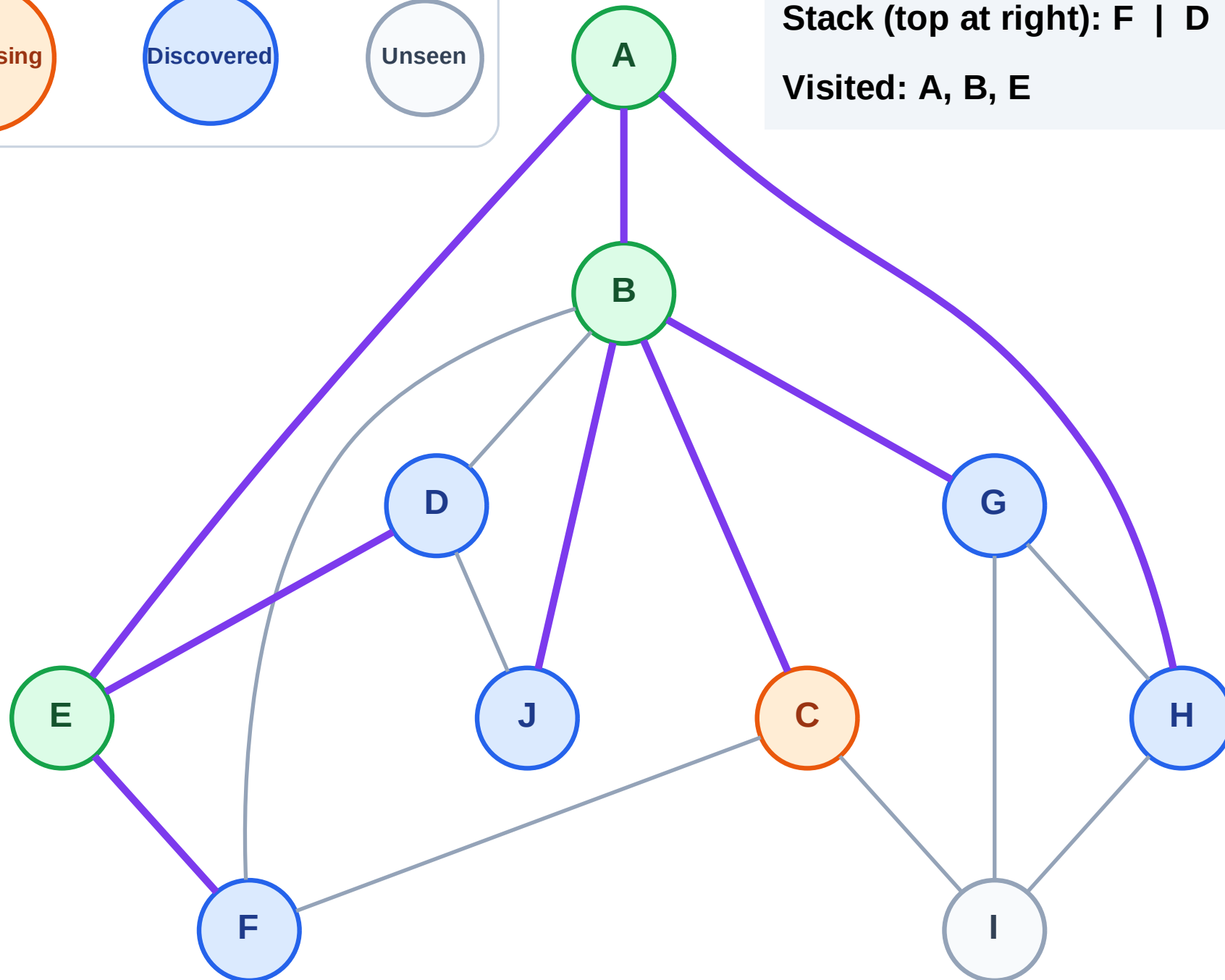
Step 8: Process node C

Legend



Stack (top at right): F | D | H | J | G

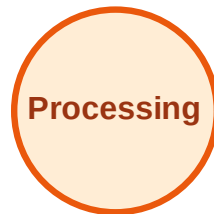
Visited: A, B, E



DFS Traversal

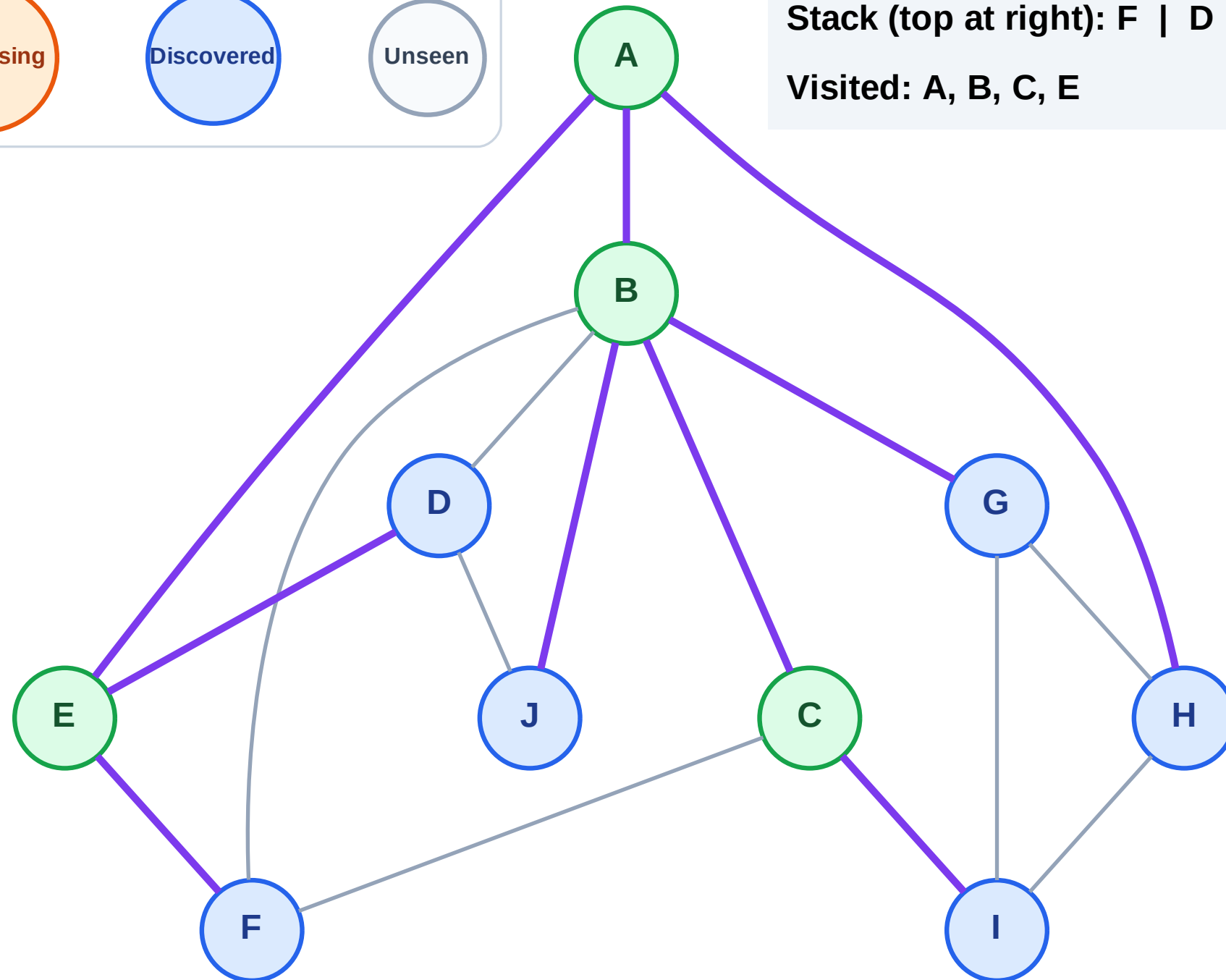
Step 9: Discover I from C

Legend



Stack (top at right): F | D | H | J | G | I

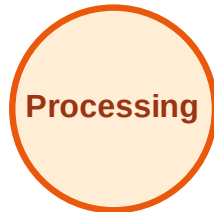
Visited: A, B, C, E



DFS Traversal

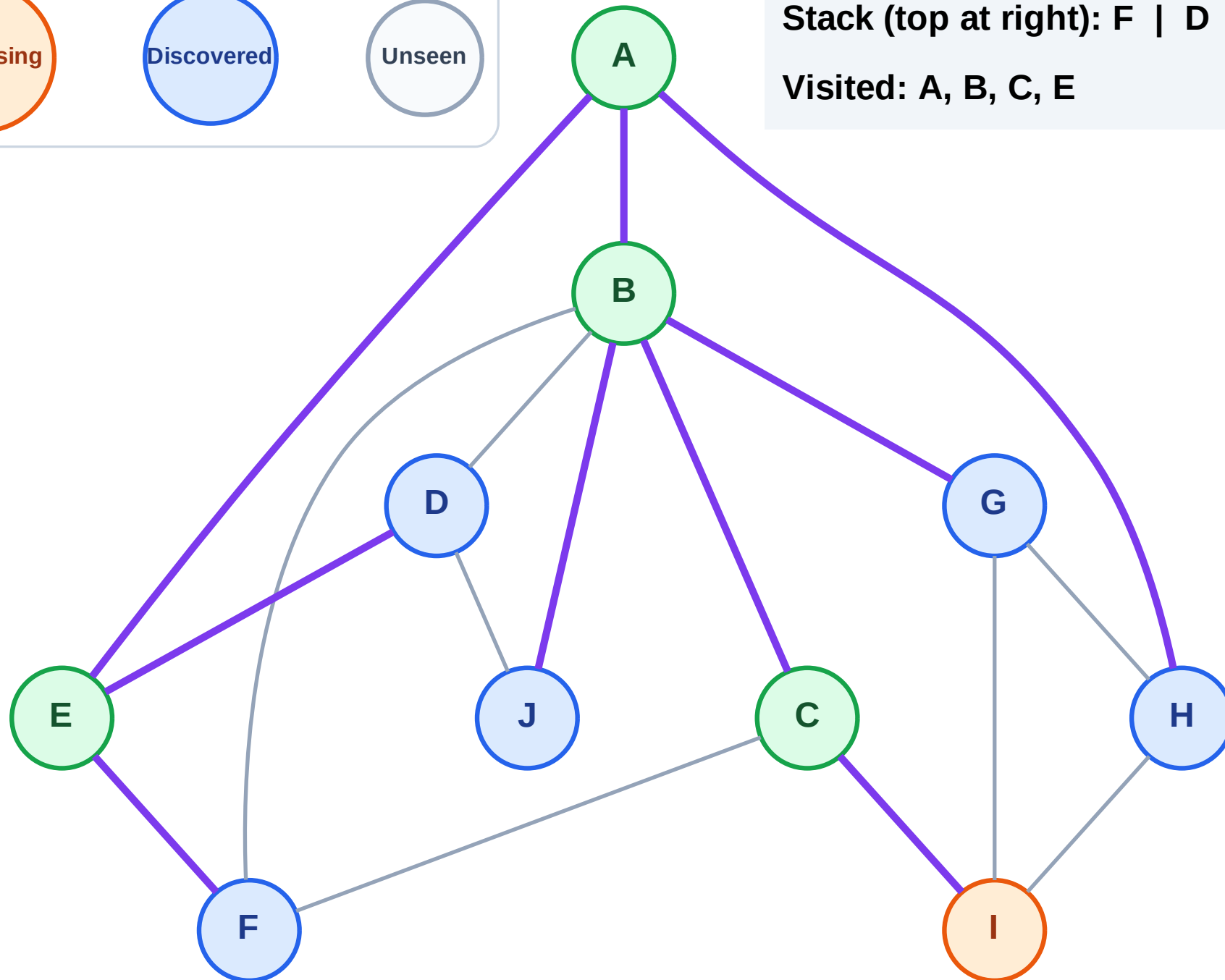
Step 10: Process node I

Legend



Stack (top at right): F | D | H | J | G

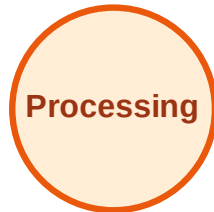
Visited: A, B, C, E



DFS Traversal

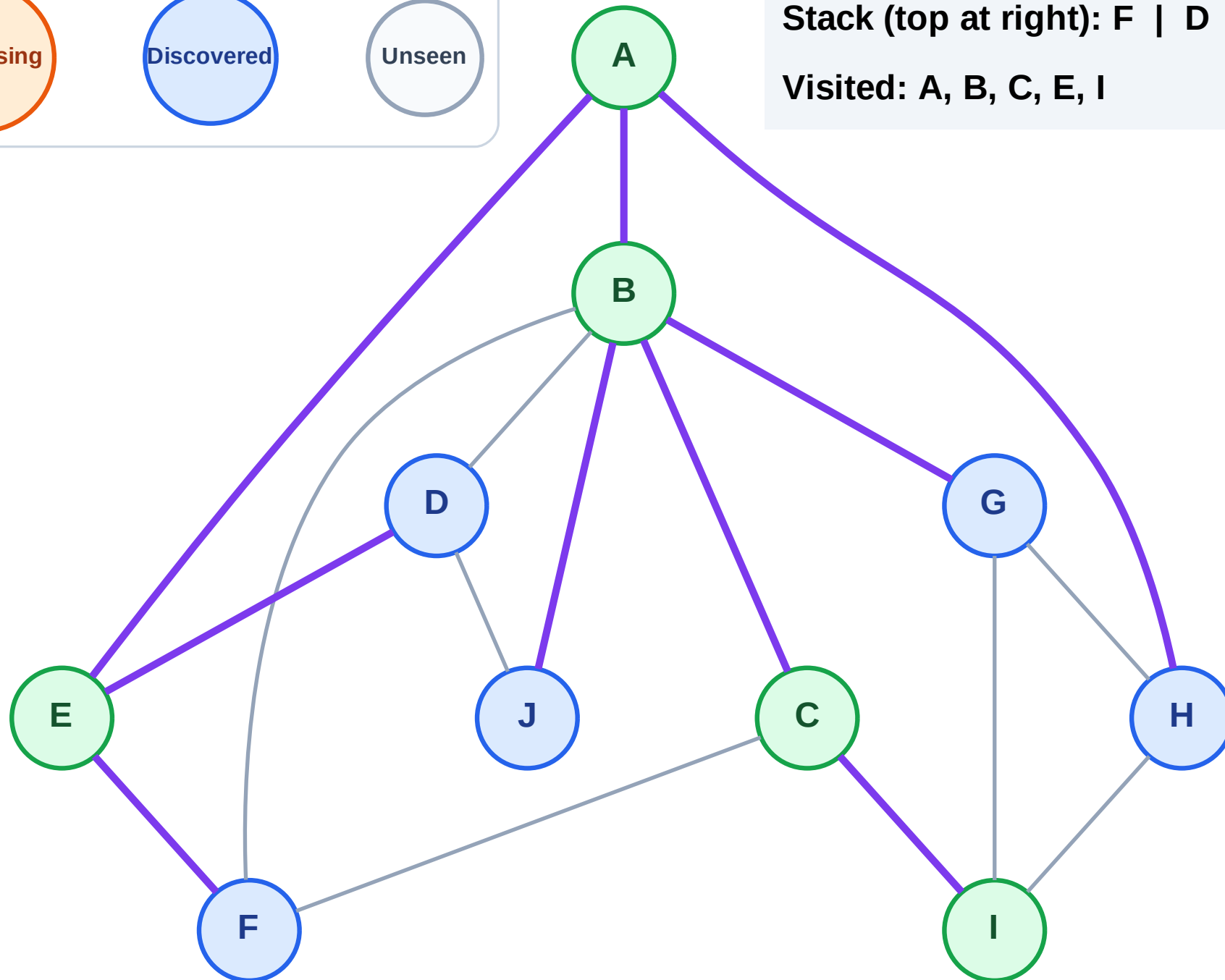
Step 11: No new neighbors from I

Legend



Stack (top at right): F | D | H | J | G

Visited: A, B, C, E, I



DFS Traversal

Step 12: Process node G

Legend

Complete

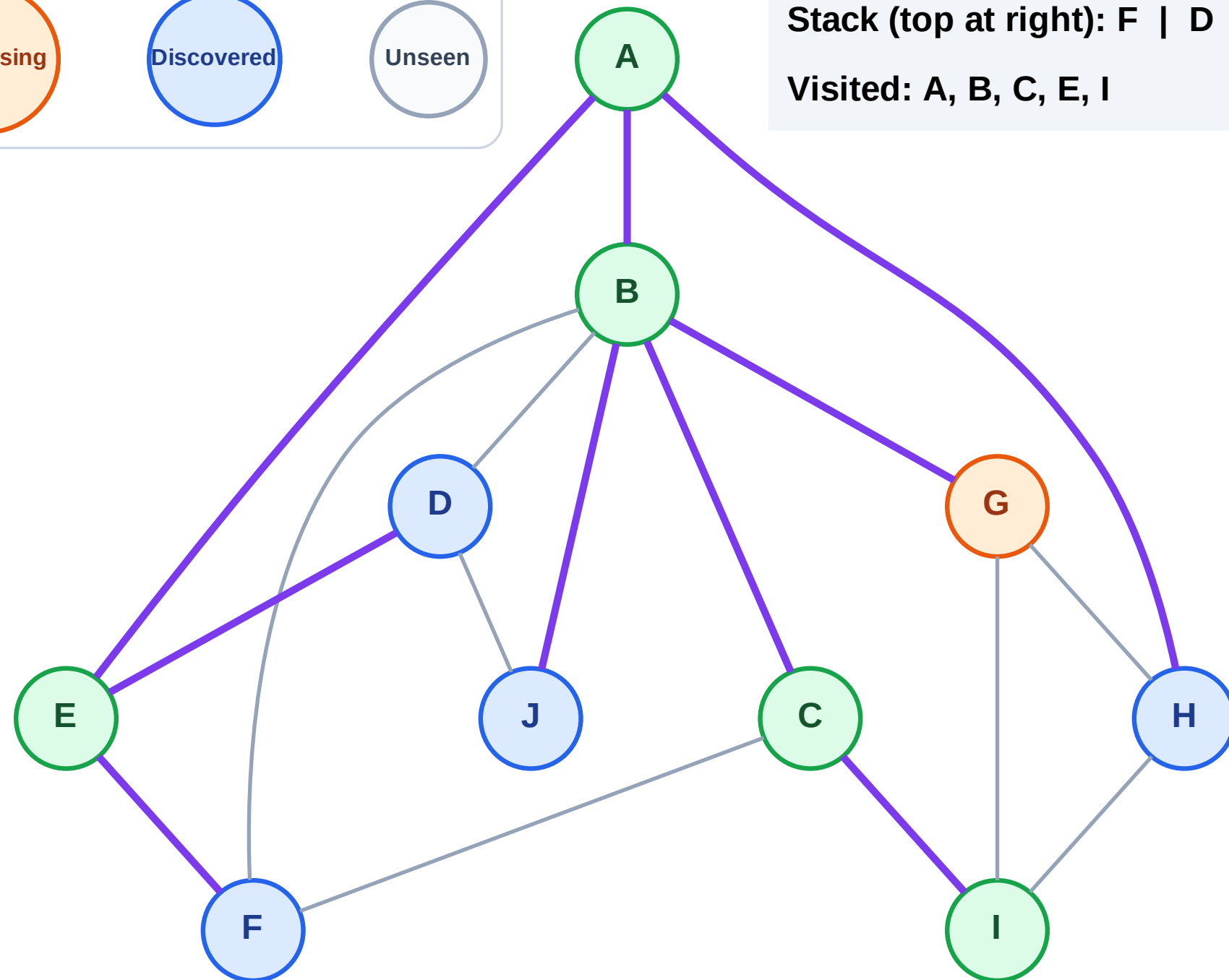
Processing

Discovered

Unseen

Stack (top at right): F | D | H | J

Visited: A, B, C, E, I



DFS Traversal

Step 13: No new neighbors from G

Legend

Complete

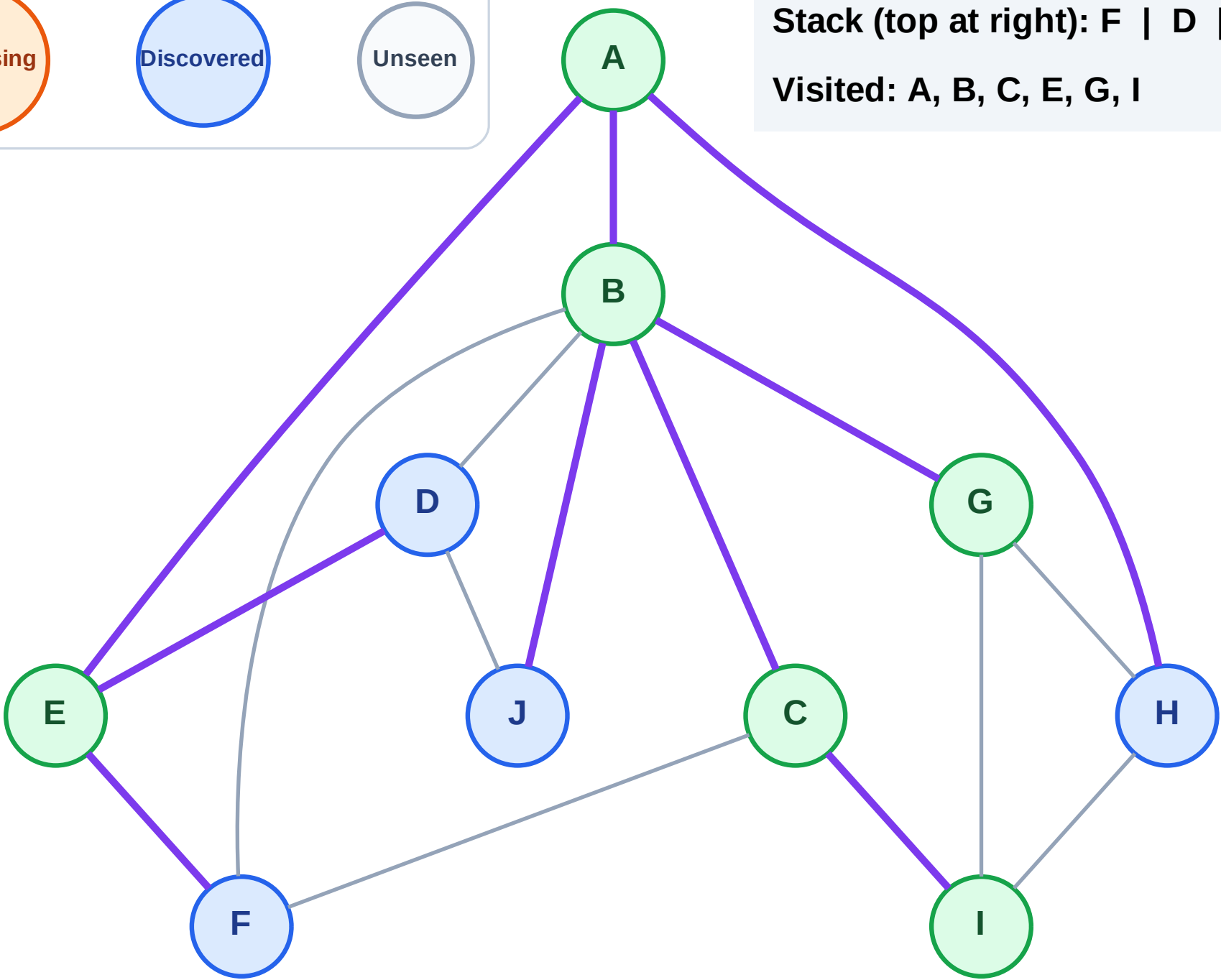
Processing

Discovered

Unseen

Stack (top at right): F | D | H | J

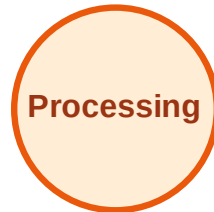
Visited: A, B, C, E, G, I



DFS Traversal

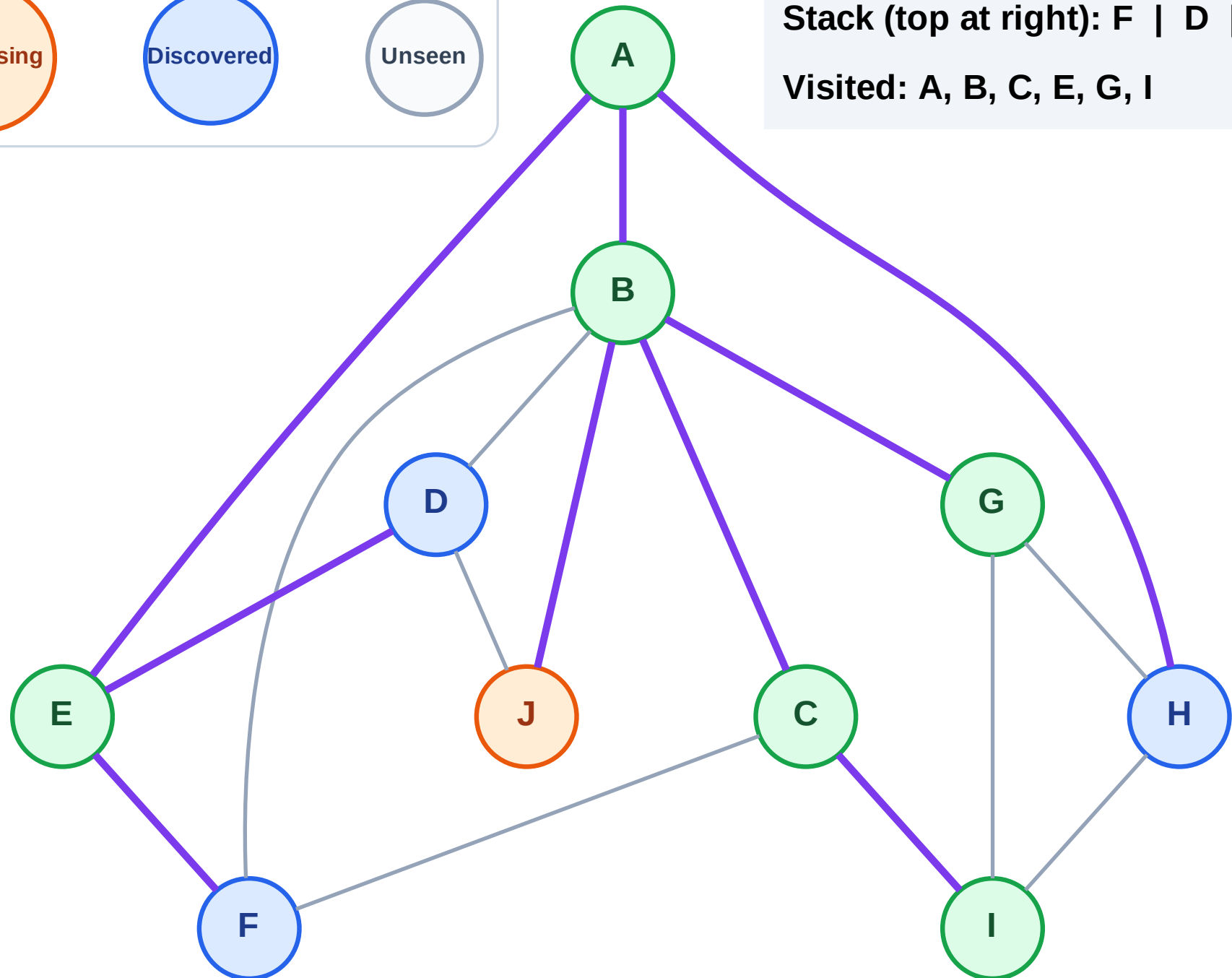
Step 14: Process node J

Legend



Stack (top at right): F | D | H

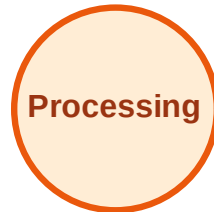
Visited: A, B, C, E, G, I



DFS Traversal

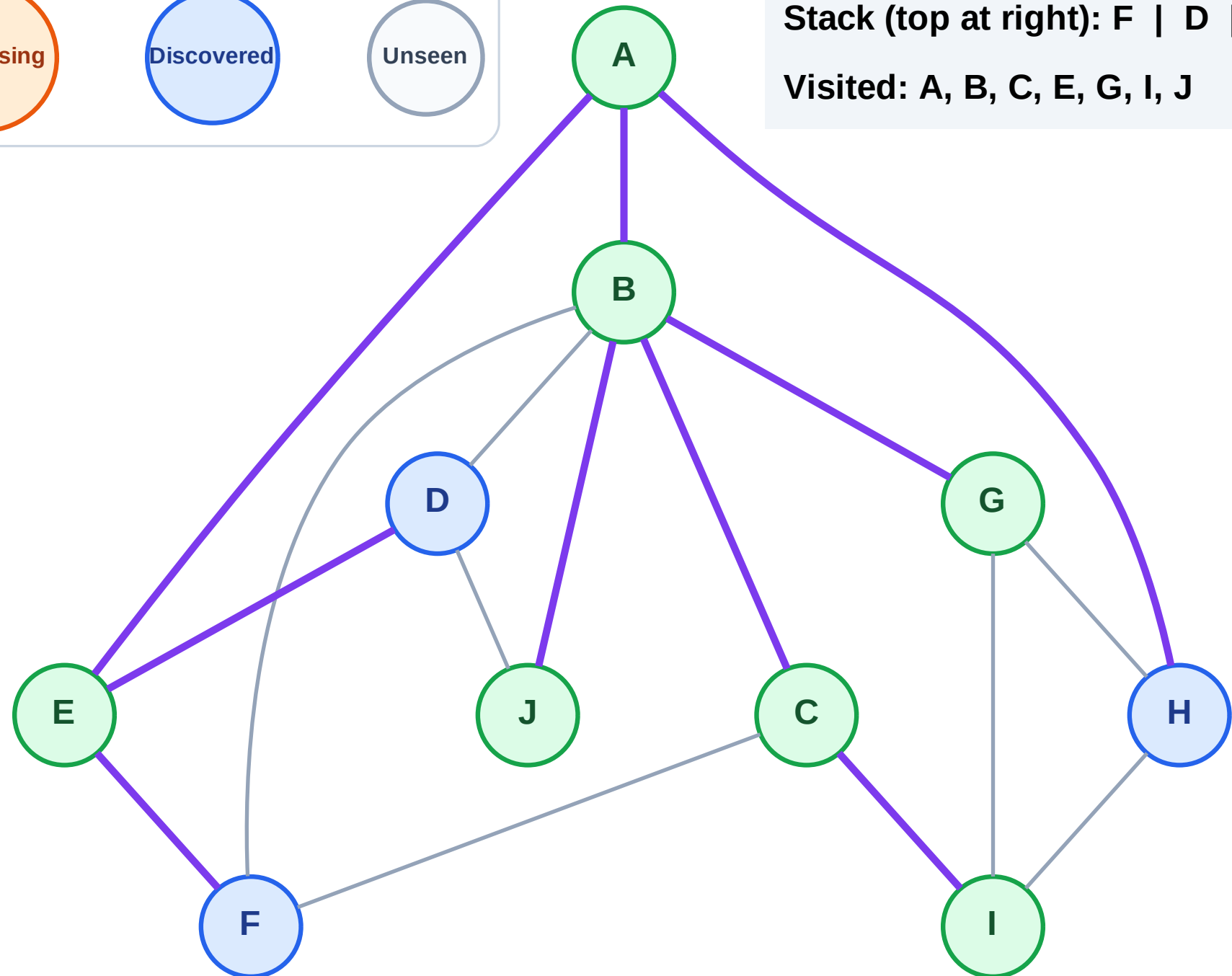
Step 15: No new neighbors from J

Legend



Stack (top at right): F | D | H

Visited: A, B, C, E, G, I, J



DFS Traversal

Step 16: Process node H

Legend

Complete

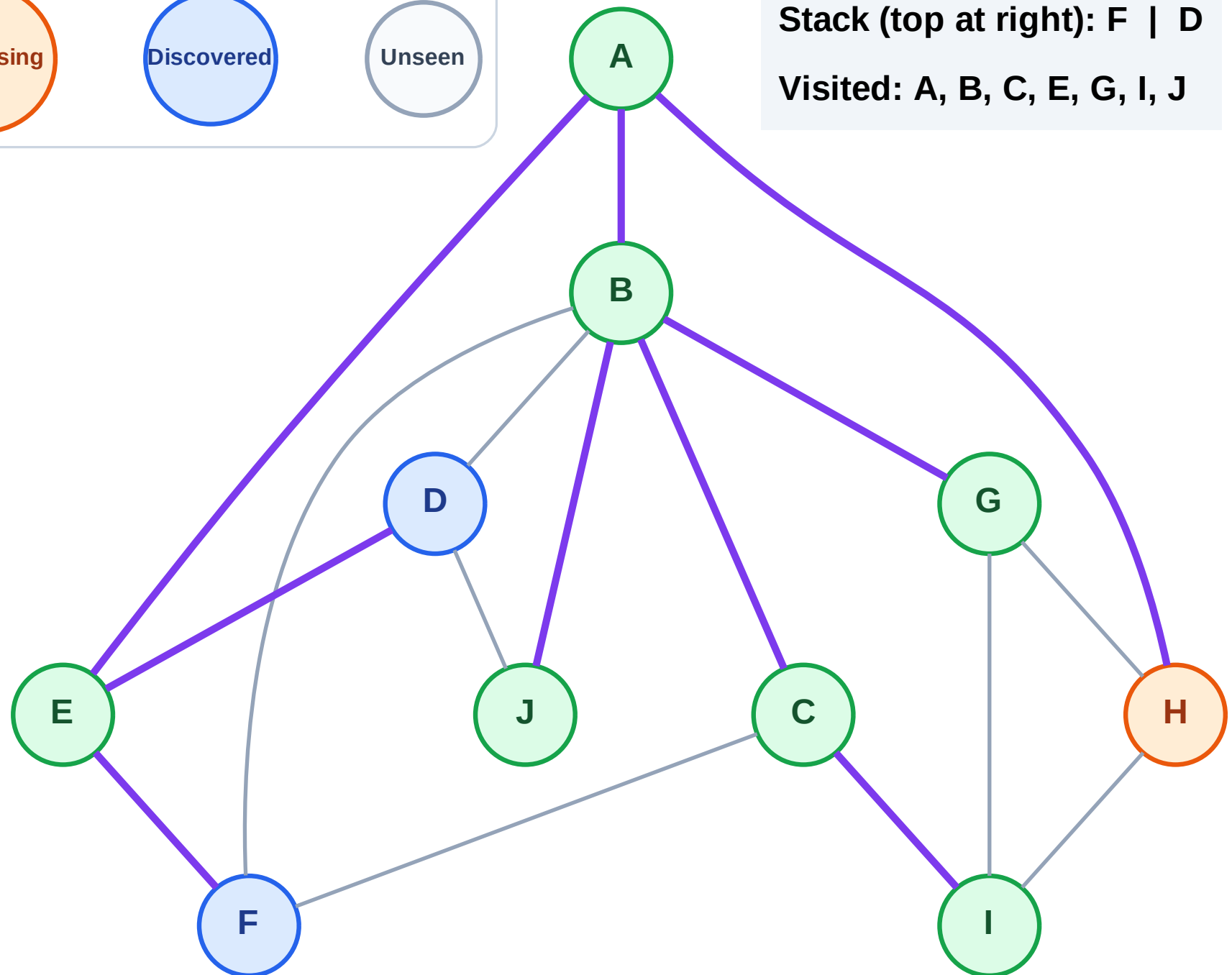
Processing

Discovered

Unseen

Stack (top at right): F | D

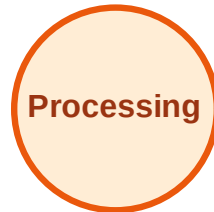
Visited: A, B, C, E, G, I, J



DFS Traversal

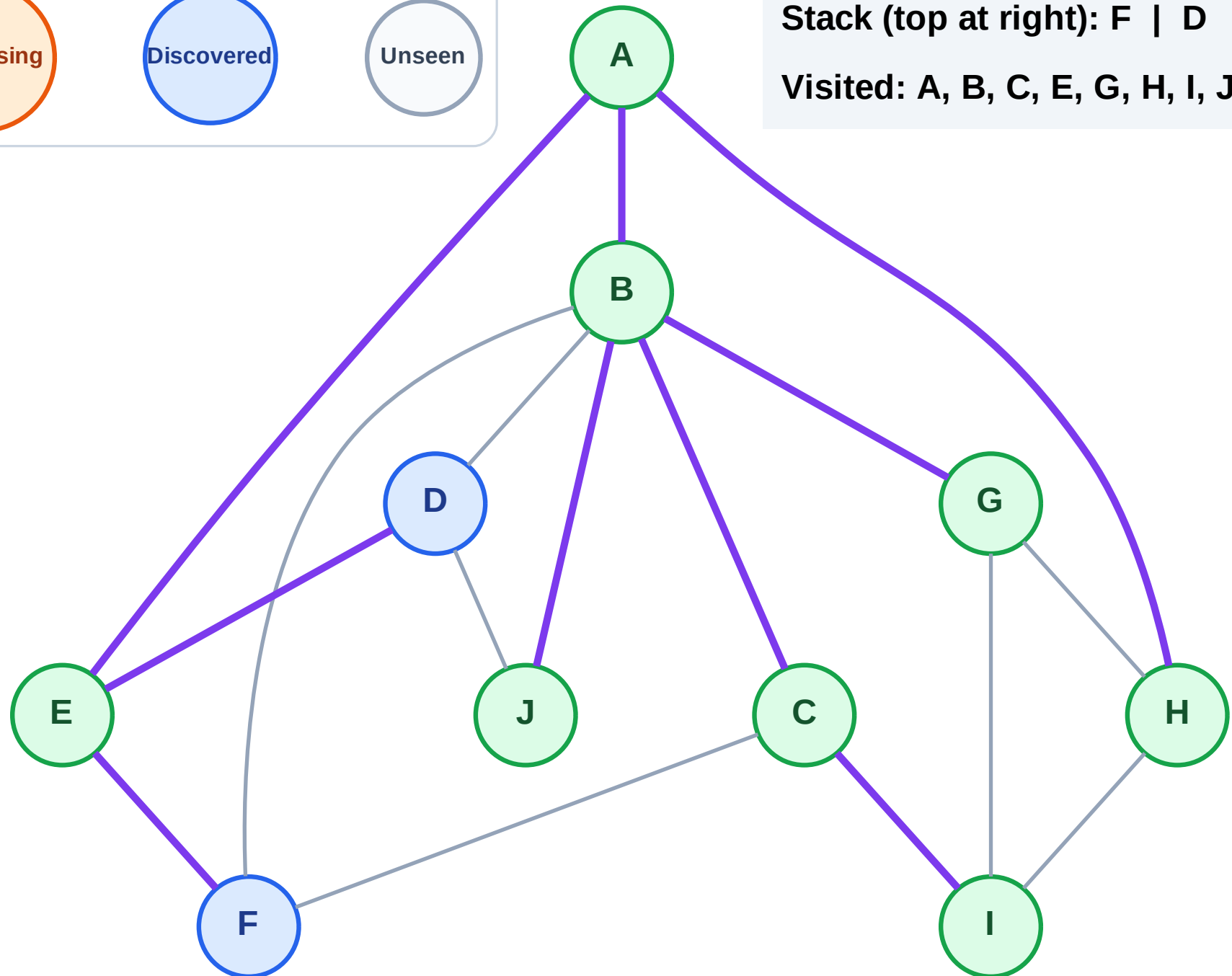
Step 17: No new neighbors from H

Legend

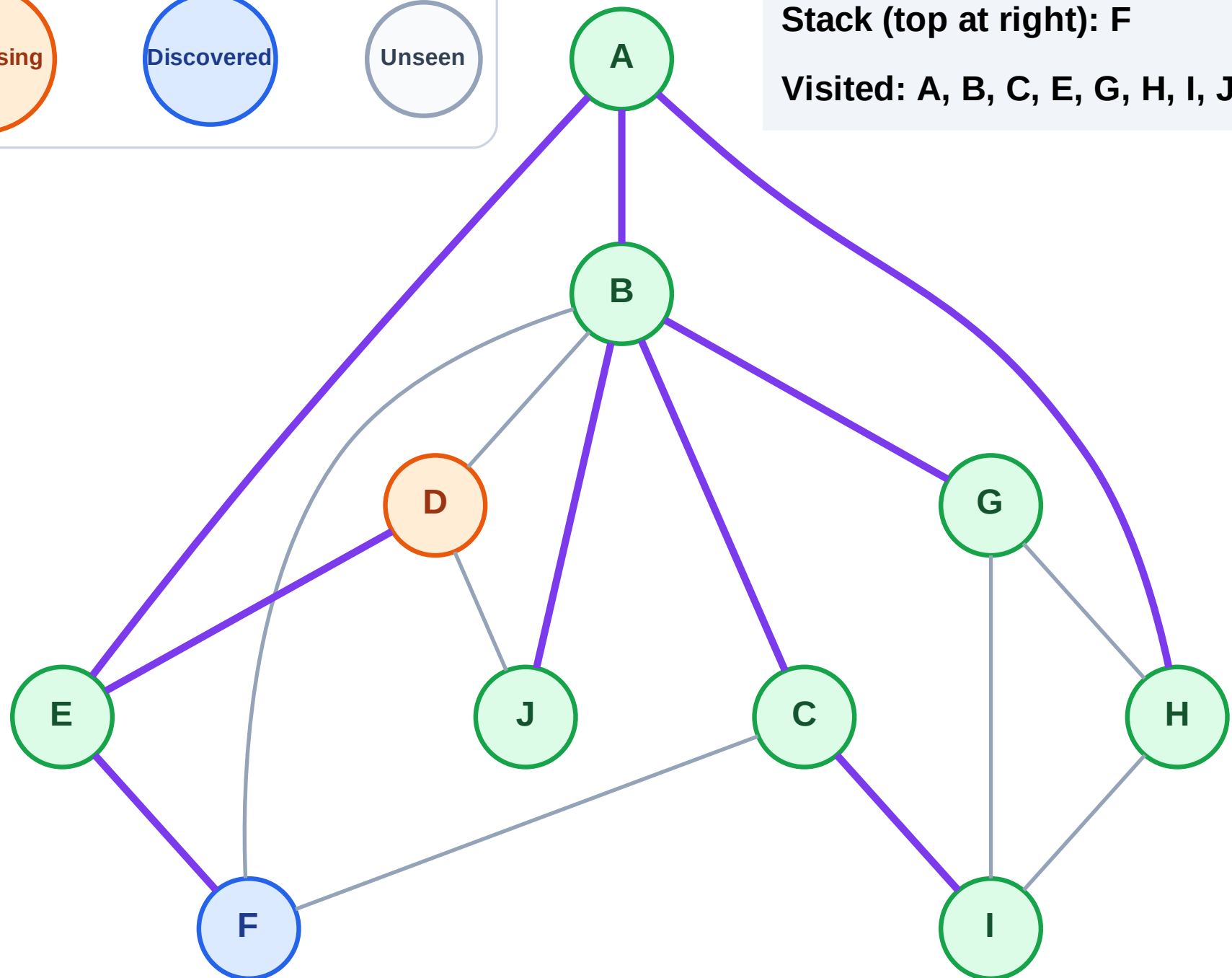
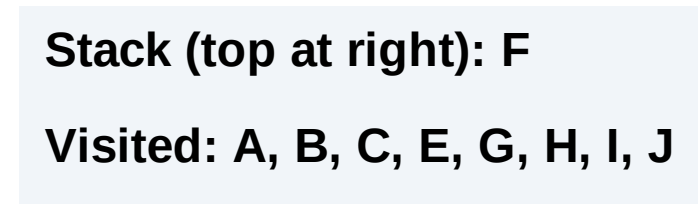


Stack (top at right): F | D

Visited: A, B, C, E, G, H, I, J



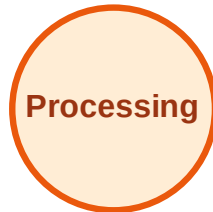
Step 18: Process node D



DFS Traversal

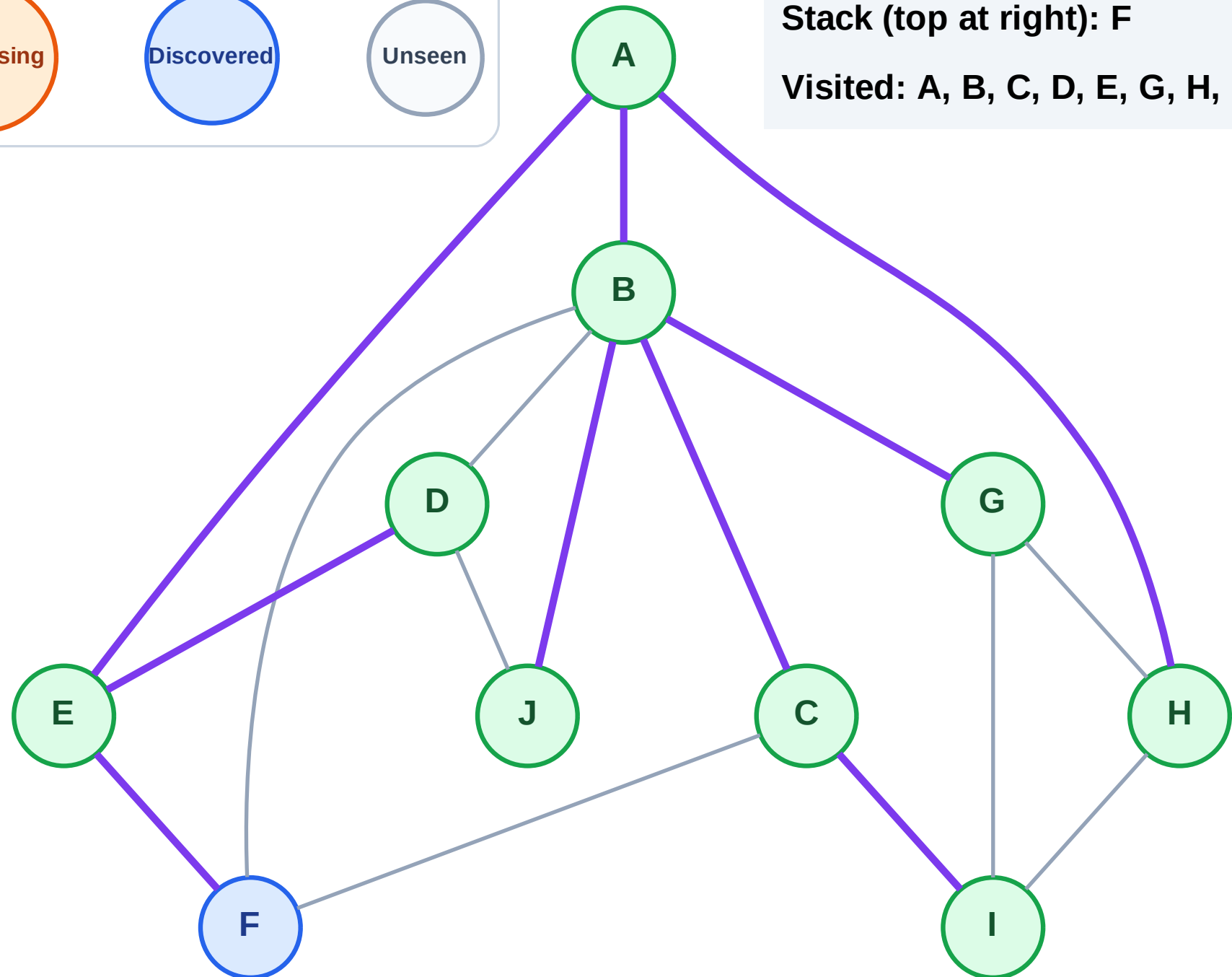
Step 19: No new neighbors from D

Legend



Stack (top at right): F

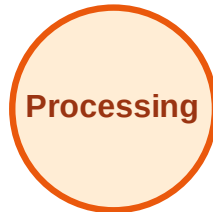
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

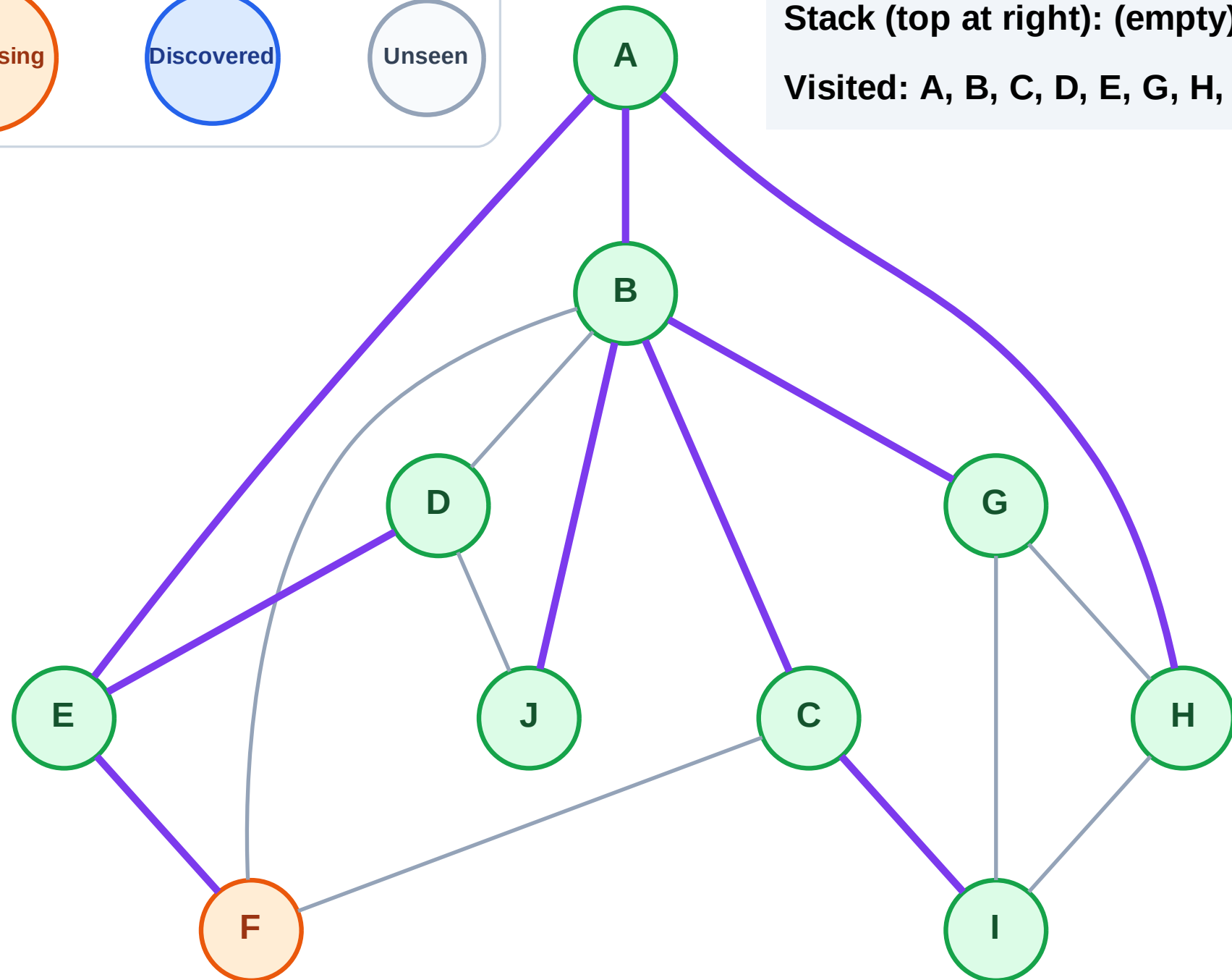
Step 20: Process node F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

